

CHAPTER SIX

BIOLOGICAL NITROGEN REMOVAL

by

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1. PROCESS FUNDAMENTALS

In this chapter the kinetics of the single sludge nitrification-denitrification process will be considered in detail. It will be shown that for municipal wastewaters, significant or even complete, removal of nitrate is possible with this process.

There are two main biological activities whereby nitrogen is removed from the wastewater: (1) sludge production and (2) respiration for energy abstraction from organic compounds. These two reactions differ fundamentally in their biological objectives.

1.1 Nitrogen removal by sludge production

The nitrogen removal attributable to sludge production was considered in detail in Chapter 4, Section 9, and it was concluded that for most municipal wastewaters only a minor fraction of nitrogen could be removed by sludge production (see Figure 4.5).

1.2 Biological denitrification

This biological reaction is known as dissimilative reduction of nitrogen or denitrification and involves the reduction of nitrate or nitrite present in the wastewater to gaseous nitrogen which escapes to the atmosphere. Removal of nitrogen via this process is a consequence of biological redox reactions, to obtain energy from the organic material. In these reactions nitrate and nitrite serve the same function as oxygen, i.e. that of a hydrogen ion/electron acceptor (see Chapter 1).

Nitrate readily replaces oxygen as the terminal electron acceptor because the electron pathway is very similar to that of oxygen. The difference is in the electron transfer from the cytochromes where the specific enzyme reductase is replaced by the enzyme nitrate reductase which catalyses the final electron transfer to nitrate instead of oxygen (see Chapter 1). Studies of pure cultures of denitrifying organisms indicate that the presence of dissolved oxygen prevents the formation of the enzyme necessary for the final electron transfer to nitrate (Chang and Morris, 1972). (See Section 2.2 below).

From stoichiometry, it can be shown that when nitrate serves as the terminal electron acceptor, the equivalent mass of oxygen, (as O) if oxygen were the terminal electron acceptor, is

$$1 \text{ mgNO}_3 \text{ as N} \equiv 2,86 \text{ mgO as O} \quad (6.1)$$

Knowing the equivalence between nitrate and oxygen, nitrate consumption for synthesis of new organism mass can be formulated with the equations for oxygen consumption by only slight modification. It also allows

COD balances to be properly executed on denitrification processes.

1.3 Effect of denitrification on oxygen demand

In Chapter 5, Section 1, it was shown that when 1 mgN ammonia or TKN is converted to nitrate 4,57 mgO are required. In the section above it was shown that when 1 mgN nitrate is denitrified it is equivalent to 2,86 mgO being supplied. Thus for nitrification, 4,57 mgO/mgN are required, but in denitrification 2,86 mgO/mgN can be recovered, i.e. with denitrification $2,86/4,57 \cdot 100 = 63\%$ of the oxygen demand for nitrification can be recovered, if complete denitrification is achieved. Because the oxygen requirement for nitrification is about 25 to 35% of the total oxygen requirement, a saving of 15 to 20% of the total oxygen requirement can be made by including biological denitrification. This is demonstrated in Figure 6.1 where the oxygen demand for (i) carbonaceous, (ii) total including nitrification and (iii) total including nitrification and denitrification (per kgCOD load) is plotted versus sludge age for a typical raw wastewater (see Tables 4.3, 5.2 and 6.1).

Other advantages of processes including denitrification are discussed in Chapter 4, Section 11.2; see also Chapter 5, Section 4.3 and Chapter 6, Section 1.4.

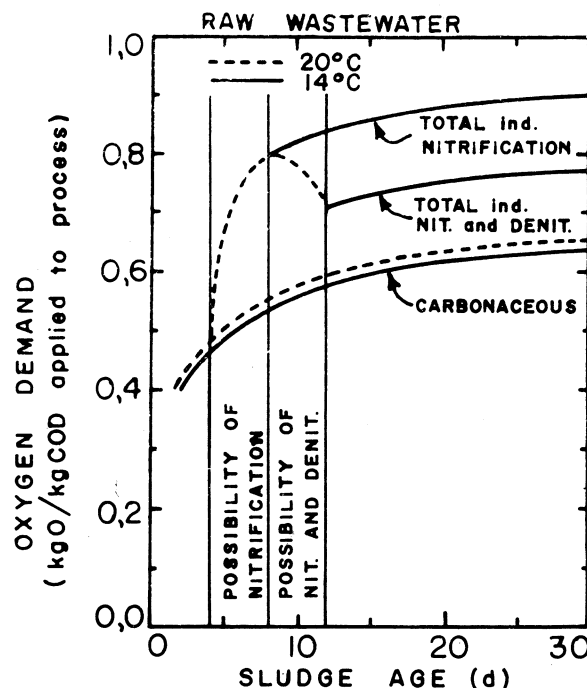


FIGURE 6.1: Carbonaceous, total including nitrification and total including nitrification and denitrification oxygen demand per unit COD load versus sludge age for raw wastewater.

1.4 Effect of denitrification on Alkalinity

Biological denitrification in which nitrate is reduced to nitrogen gas is accompanied by an increase in Alkalinity. From stoichiometry, for 1 mgN nitrate denitrified, 3.57 mg as CaCO_3 Alkalinity is produced. In Chapter 5, Section 4.3 it was shown that in the nitrification process, 7.14 mg/l as CaCO_3 alkalinity was lost for every 1 mgN/l ammonia nitrified. Consequently, through denitrification, half the decrease in alkalinity due to nitrification can be recovered.

The recovery of alkalinity by denitrification is important when treating wastewaters with low alkalinity values – less than 200 mg/l as CaCO_3 . For such wastewaters the production of only 15 mgN/l nitrate without denitrification will reduce the mixed liquor pH value to below 5. In this event, because nitrification is extremely sensitive to pH values below 7, only partial nitrification will be obtained, even at long sludge ages. However, should denitrification be incorporated in the process, the recovery of alkalinity by denitrification usually will cause the mixed liquor pH value to be maintained near 7 even for low alkalinity wastewaters. This favourable effect, of maintaining the mixed liquor pH value near 7, makes a nitrification – denitrification process virtually mandatory for activated sludge processes treating wastewaters with low alkalinity values. This aspect is discussed in detail in Chapter 5, Section 4.3 and Chapter 4, Section 11.2.

2. REQUIREMENTS FOR DENITRIFICATION

For denitrification to occur, the following conditions are necessary:

- (1) Presence of nitrate (or nitrite)
- (2) Absence of dissolved oxygen
- (3) A facultative bacterial mass
- (4) Presence of a suitable electron donor (energy source).

2.1 Presence of nitrate

Normally the presence of nitrate implies nitrification as a prerequisite. The conditions for adequate nitrification in an activated sludge process have been discussed in detail in Chapter 5, Section 4. The prerequisite for nitrification involves the estimation of the maximum sludge mass fraction which may be left unaerated but still ensuring nitrification.

2.2 Absence of dissolved oxygen

The inhibitory effect of dissolved oxygen on denitrification has been extensively reported in the literature. In a comparative test series, Carlson (1972) found that at zero dissolved oxygen level the nitrate removed was 100 per cent, while at 0.2 mg/l dissolved oxygen (2 to 10 per cent oxygen saturation), no significant denitrification was obtained. Furthermore, it should be remembered that in a completely mixed anoxic reactor, even if no measurable dissolved oxygen concentration (DO) is present, any oxygen entering such a reactor will be utilized preferentially by the organism mass, thereby

reducing (in stoichiometric proportion) the mass of nitrate the reactor can denitrify. Possible sources of DO are: (1) high DO concentrations in the recycles from the aerobic zones to the anoxic zones – the DO concentration should be controlled to be between 1 and 2 mgO/l i.e. sufficiently high so as not to inhibit nitrification in the aerobic reactor but not too high to reduce significantly the denitrification in the anoxic reactor; (2) entrainment into the anoxic reactor via the air-water interface due to unnecessarily high mixing intensities; (3) Archimedian screw pumps for recycle pumping; (4) hydraulic jumps, cascades and shooting flow conditions in interconnecting channels – closed channels and tranquil flow conditions entrain less oxygen. (See also Chapter 10 on operation of nutrient removal plants).

2.3 Facultative bacterial mass

The propensity to denitrify is widespread among bacteria. Dissimilative denitrification with end products N_2 , NO and NO_2 has been established in numerous cases. The bulk of the bacterial mass in wastewater treatment is facultative and a significant fraction is capable of dissimilative denitrification. Studies on denitrification tend to support the view that there is little difference between the bacterial masses in processes where nitrification only or nitrification-denitrification takes place. There is little merit therefore in attempting to analyse the bacterial composition of the sludge in detail. A sludge generated under aerobic conditions, when subjected to the appropriate environmental conditions, will show a denitrifying capability immediately and will continue to do so, subsequently without apparent change in reactivity.

2.4 Electron donor (Energy source)

The oxidation of carbonaceous compounds with nitrate as electron acceptor i.e. denitrification, provides the energy required by the facultative heterotrophs for synthesis of new mass and endogenous respiration. The electron donor constitutes the source of the energy. The axis about which all biological denitrification investigations revolve is the energy source which serves as the electron donor in the denitrification process. Denitrification has been investigated for a variety of substances as carbonaceous energy sources. These can be categorised as follows:

- (a) Energy sources not present in the wastewater, i.e. *an external carbonaceous energy source* which is added at the denitrification stage of the process. Compounds used as an external energy source include methanol, methane, ethanol, acetone and acetic acid.
- (b) Energy sources present in the influent wastewater, i.e. *an internal carbonaceous energy source* which enters the system with the wastewater.
- (c) Energy sources which are *self-generated* within the system by the release of nutrient by the organisms in the death phase.

The type of energy source utilised has led to different process configurations for denitrification studies. Usually in municipal wastewater treatment only energy sources (b) and (c) above are of interest as these are sources of energy available in the single sludge activated sludge process. However, in some nutrient removal plants a readily biodegradable COD source may be added to the anaerobic reactor to improve the biological P removal (see Chapter 4, Section 8 and Chapter 7, Section 3.3). This practice not only improves P removal but also the removal of nitrate by denitrification as it constitutes the addition of an external energy source (group a). Sources of readily biodegradable COD are acid digester supernatant or some form of organic wastewater with a large readily biodegradable COD fraction such as malting industry wastewater, fruit and vegetable canning factory wastewaters, juicing factory wastewaters, etc. The addition of these wastes increases the nitrate removal potential by denitrification because the additional readily biodegradable COD allows the initial rapid rate of denitrification to operate for a longer period of time (see Section 4.1.1 below).

3. DENITRIFICATION PROCESS CONFIGURATION

Two basic activated sludge process configurations have been developed for single sludge biological denitrification: (1) the Wuhrmann configuration and (2) the Ludzack-Ettinger configuration.

3.1 The Wuhrmann configuration

The single sludge nitrification-denitrification system in which endogenous energy release provides the energy source for denitrification was first proposed by Wuhrmann (1964). A schematic presentation of the process is shown in Figure 6.2. It consists of two reactors in series, the first aerobic and the second anoxic. The influent is discharged to the first reactor where aerobic growth of both the heterotrophic and nitrifying organisms takes place. Provided the sludge age is sufficiently long and the aerobic fraction of the system is

adequately large, nitrification will be complete in the first reactor.

The mixed liquor from the aerobic reactor passes to the anoxic reactor, also called the *post-denitrification reactor* or *secondary anoxic reactor* where it is kept completely mixed by stirring, but with no aeration. The outflow from the anoxic reactor passes through a settling tank and the underflow is recycled back to the aerobic reactor.

Energy release by the sludge mass due to the death of organisms provides the energy source for denitrification in the anoxic reactor. However, the rate of release of energy is low, so that the rate of denitrification is also low. Consequently, in order to obtain a meaningful reduction of the nitrate concentration in the anoxic reactor, the anoxic fraction of the system must be large and this may cause a breakdown of the nitrification process. Thus, although theoretically the system has the potential to remove all the nitrate, from a practical point this usually is not possible as the anoxic volume fraction will need to be so large that the conditions for nitrification cannot be satisfied particularly if the temperatures are low, below 15°C. Furthermore, in the anoxic reactor, organic nitrogen and ammonia are released through organism death, some of which passes out with the effluent thereby reducing the total nitrogen removal of the system. To minimise the ammonia content of the effluent, a flash or reaeration reactor may be placed between the anoxic reactor and the settling tank. In this reactor the ammonia is then nitrified to nitrate thereby reducing overall efficiency of the nitrate reduction capability of the process.

3.2 The Ludzack-Ettinger configuration

Ludzack and Ettinger (1962) were the first to propose a single sludge nitrification-denitrification process utilizing the biodegradable material in the influent as an energy source for denitrification. A schematic presentation of this system is shown in Figure 6.3. It consists of two reactors in series, *partially* separated from each other. The influent is discharged to the first reactor which is maintained in an anoxic state by stirring without aeration. The second reactor is aerated and

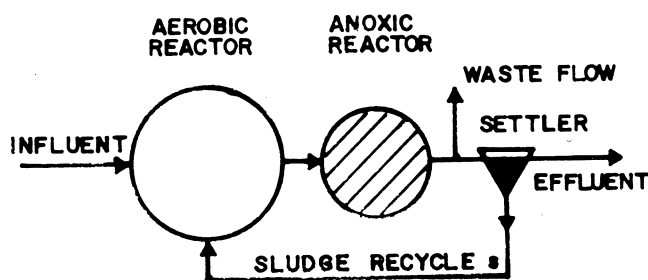


FIGURE 6.2: The Wuhrmann process for nitrogen removal.

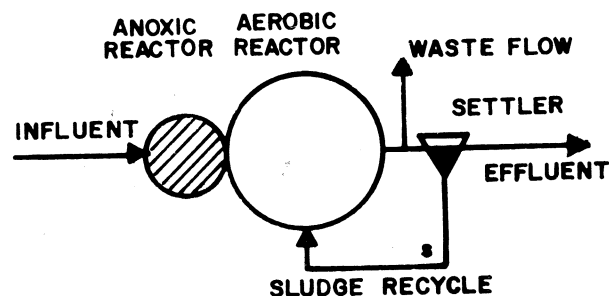


FIGURE 6.3: The Ludzack-Ettinger process for nitrogen removal.

nitrification takes place. As there is only partial separation, the mixed liquor in the first reactor is in communication with that in the second reactor. Due to the mixing action in both reactors, an interchange of the nitrified and anoxic liquors is induced, and the nitrate entering the anoxic reactor is reduced to nitrogen gas. Ludzack and Ettinger reported that the process gave variable denitrification results, probably due to the lack of control of the interchange of the contents of the two reactors.

In 1973, Barnard in developing the Bardenpho process proposed an improvement of the Ludzack-Ettinger process, by completely separating the anoxic and aerobic reactors, recycling the underflow from the settler to the anoxic reactor, and providing an additional recycle from the aerobic to the anoxic reactor (Figure 6.4).

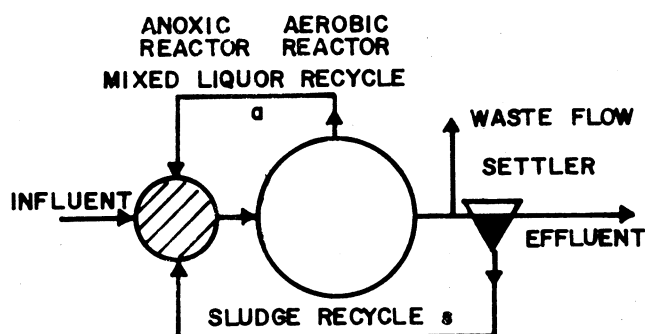


FIGURE 6.4: The modified Ludzack-Ettinger process for nitrogen removal.

These modifications allow a significant improvement in control over the process performance. The high influent energy source discharged to the anoxic reactor also called the *pre-denitrification reactor* or *primary anoxic reactor*, gives rise to a high rate of denitrification and a substantially higher reduction of nitrate than in the Wuhrmann process, even when the pre-denitrification reactor of this process is substantially smaller than the post-denitrification reactor of the Wuhrmann process. For the purposes of identification, this process will be called the modified Ludzack-Ettinger process.

With the modified Ludzack-Ettinger process, complete denitrification cannot be achieved because a part of the total flow from the aerobic reactor is not recycled to the anoxic reactor but is discharged directly with the effluent.

3.3 The Bardenpho configuration

In order to overcome the deficiency of incomplete denitrification in the modified Ludzack-Ettinger process, Barnard (1973) proposed combining this process with that of Wuhrmann and called it the Bardenpho process. A schematic presentation of the Bardenpho process is shown in Figure 6.5. Barnard considered that the low concentration of nitrate discharged from the aerobic

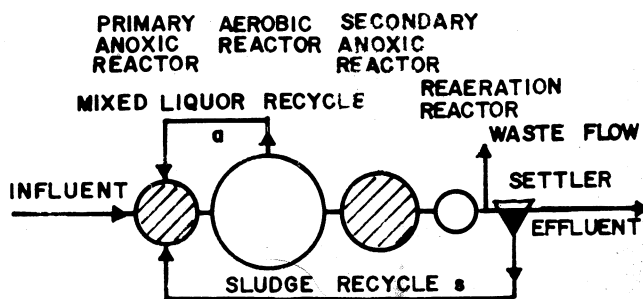


FIGURE 6.5: The Bardenpho process for nitrogen removal.

reactor to the secondary anoxic reactor will be denitrified to produce a relatively nitrate-free effluent. To strip the nitrogen bubbles generated in the secondary anoxic reactor attached to the sludge flocs, he introduced a flash aeration reactor between the secondary anoxic and the final settling tank. The flash aeration reactor was also considered necessary to nitrify the ammonia released during the sludge residence time in the secondary anoxic reactor. In order to reduce the possibility of flotation of sludge in the settler due to denitrification of residual nitrate, the sludge accumulation in the settler was to be kept to a minimum. This was achieved by having a high recycle rate from the settler, approximately equal to the mean influent flow.

Although in concept the Bardenpho process has the potential for complete removal of nitrate, in practice this is not always possible, an aspect that will be discussed in detail in Section 6 below.

4. EXPERIMENTAL BASIS FOR DENITRIFICATION KINETICS

Carlson (1971) and Christensen and Harremoës (1977) suggested that the kinetic reaction for denitrification by activated sludge mixed liquor can be expressed by

$$dN_n/dt = KX \quad (6.2)$$

where

dN_n/dt = denitrification rate (mgN nitrate/litre/unit time)

N_n = nitrate concentration (mgN/l)

t = time (hours or days)

X = volatile solids concentration (mgVSS/l)

K = specific denitrification constant (mgN/mgVSS/time).

This expression for the denitrification rate was developed from batch tests. It indicates that the nitrate-time relationship is linear and independent of the nitrate concentration, i.e. the rate is zero order with respect to nitrate concentration, and a function only of the volatile sludge concentration.

In order to evaluate Eq. 6.2 as a basis for describing denitrification behaviour in the single activated sludge process, Marais and a number of co-workers (Stern, 1974; Wilson, 1976; Marsden, 1977), undertook an ex-

tensive investigation into the kinetic behaviour of the modified Ludzack-Ettinger and Wuhrmann configurations using plug flow mixing regimes for the respective primary and secondary anoxic reactors. By measuring the nitrate concentration along the length of the plug flow anoxic reactors, the denitrification kinetic behaviour was determined by plotting nitrate concentration-time profiles. Experiments were carried out using only raw and settled wastewaters at different temperatures, sludge ages, sludge concentrations, influent COD concentrations and recycle ratios.

The investigation was divided into two parts: (1) constant flow and load conditions, and (2) cyclic flow and load conditions.

4.1 Constant flow and load conditions

4.1.1 Experimental work with plug flow reactors

Typical nitrate concentration-time profiles for secondary and primary anoxic plug flow reactors are shown in Figures 6.6 and 6.7 respectively. The secondary anoxic profiles all exhibited a single linear phase that appeared to be independent of the nitrate concentration i.e. the profiles exhibited a behavioural pattern not at variance with Eq. 6.2. The primary anoxic profiles, however, exhibited profiles that all indicated a two-stage denitrification response, i.e. (1) a fast primary phase of short duration (1 to 10 minutes), and thereafter (2) a slow second-

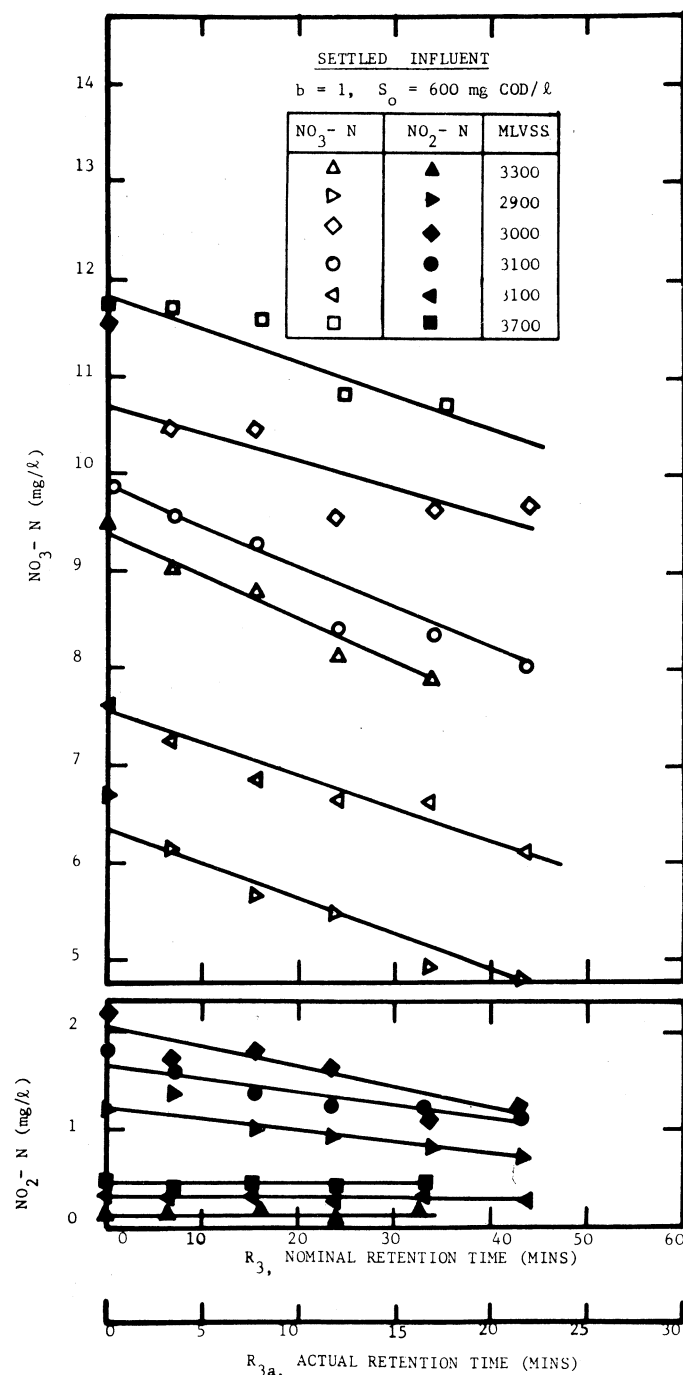


FIGURE 6.6: Nitrate concentration profiles observed in a plug-flow secondary anoxic reactor at 20°C (see Figure 6.2. After Stern and Marais, 1974).

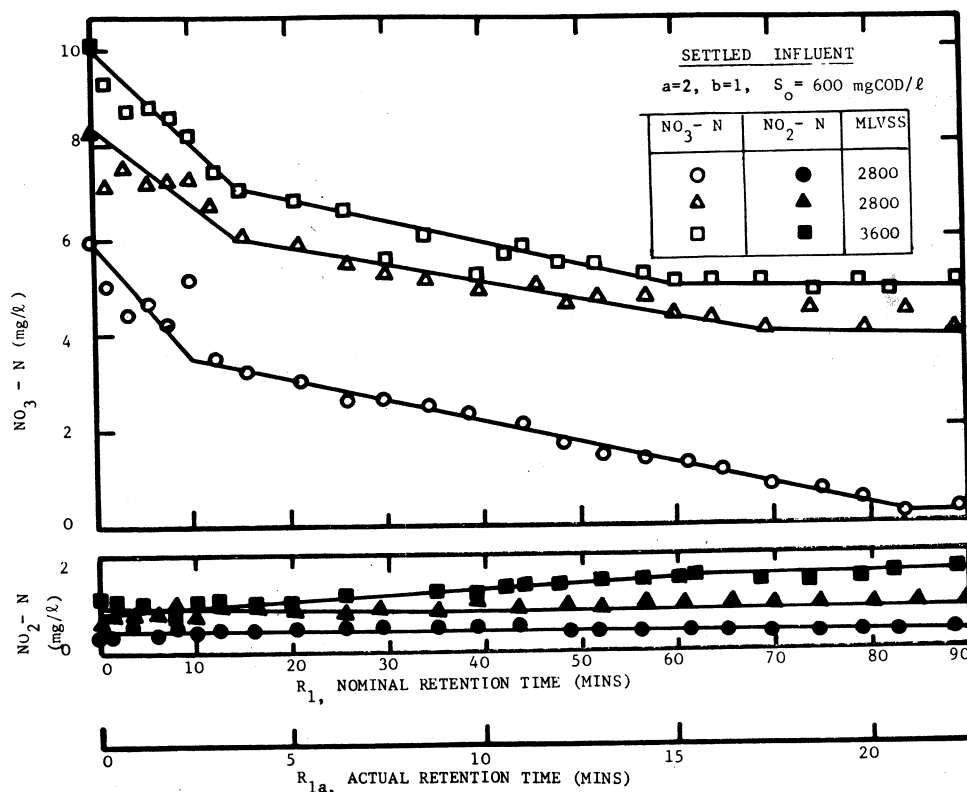


FIGURE 6.7: Nitrate concentration profiles observed in a plug-flow primary anoxic reactor at 20°C (see Figure 6.4. After Stern and Marais, 1974)

any phase at about one seventh the rate of that of the primary phase.

In order to fit the experimental data into the functional form expressed by Eq. 6.2, it is necessary to discuss the following aspects: (i) sludge concentration, (ii) denitrification rate constants, (iii) recycle ratios, and (iv) sludge age.

- (i) **Sludge concentration:** As the denitrification reaction is mediated by the active organisms, Eq 6.2 should be expressed in terms of the active mass concentration X_a , i.e.

$$\Delta N_n = K X_a \Delta t \quad (6.3)$$

In all the investigations by Marais and his co-workers it was found that the sludge production in anoxic-aerobic systems was not distinguishably different from a comparable aerobic system, provided that the anoxic volume fraction was not too great (i.e. less than about 50 per cent). Consequently, the active mass concentration X_a in the system could be expressed by Eq. 4.10 (see Chapter 4, Section 4), (Arkley and Marais, 1982).

- (ii) **Denitrification rate constants, K :** If the denitrification rate constant, K , maintains the same value throughout the actual residence time in the plug-flow reactor, R_a , then the difference between the influent and effluent nitrate concentrations, N_{ni} and N_{no} respectively, can be written in terms of Eq. 6.3

$$N_{ni} - N_{no} = \Delta N_{na} = K X_a R_a \quad (6.4)$$

where

R_a = actual retention time in plug flow reactor (h or d)

Eq. 6.4 is valid only if the effluent nitrate from the reactor is greater than zero, see Eq. 6.17 below. The difference $(N_{ni} - N_{no})$ is the *actual* removal of nitrate concentration in the *reactor*. However, as the rate of flow through the reactor is $Q(1+a)$ (where Q =units of influent flow to the plant per unit of time and a =recycle ratio), the flow passes $(a+1)/1$ times through the reactor. This gives the *system* removal, N_{ns} , i.e.

$$\Delta N_{ns} = (a+1)(N_{ni} - N_{no}) \quad (6.5)$$

The *nominal* retention time of the plug-flow reactor is defined by

$$R_n = V_r / Q \quad (6.6)$$

where

R_n = nominal retention time

V_r = volume of the reactor (l)

Q = daily average influent flow to process.

Hence the relationship between the actual and nominal retention times is

$$R_a = R_n / (1+a) \quad (6.7)$$

Substituting Eq. 6.4 for $N_{ni} - N_{no}$ and Eq 6.7 for R_a into Eq. 6.5, the system nitrate removal is given by

$$\Delta N_{ns} = K X_a R_n \quad (6.8)$$

Comparing Eqs. 6.4 and 6.8, note that K is independent of the recycle ratio a . Hence K can be determined directly from Eq. 6.4 i.e. from the actual observed reduction in the nitrate concentration in the anoxic reactor. With these basic equations the nitrate reduction in the primary and secondary anoxic reactors now can be formulated.

● Secondary anoxic reactor:

Referring to the experimental secondary anoxic reactor profiles, Figure 6.6, only one constant linear nitrate time reduction phase was observed and the analyses above allow the denitrification rate constant in this reactor, K_3 to be determined directly from Eq. 6.4 i.e.

$$\Delta N_{nss} = K_3 X_a R_{ns} \quad (6.9)$$

where the additional subscript s refers to the secondary anoxic reactor.

By applying Eq. 6.9 to all the profiles observed experimentally, a set of K values was obtained at 14°C and 20°C respectively and the mean value determined by a graphical statistical analysis. The statistical plot of the K_3 values obtained at 14°C is shown in Figure 6.8. From the mean K_3 values at 14°C and 20°C the effect of temperature on K_3 was formulated as

$$K_{3T} = 0,072 (1,03)^{T-20} \text{ (mgNO}_3\text{-N/mgVASS/d)} \quad (6.10)$$

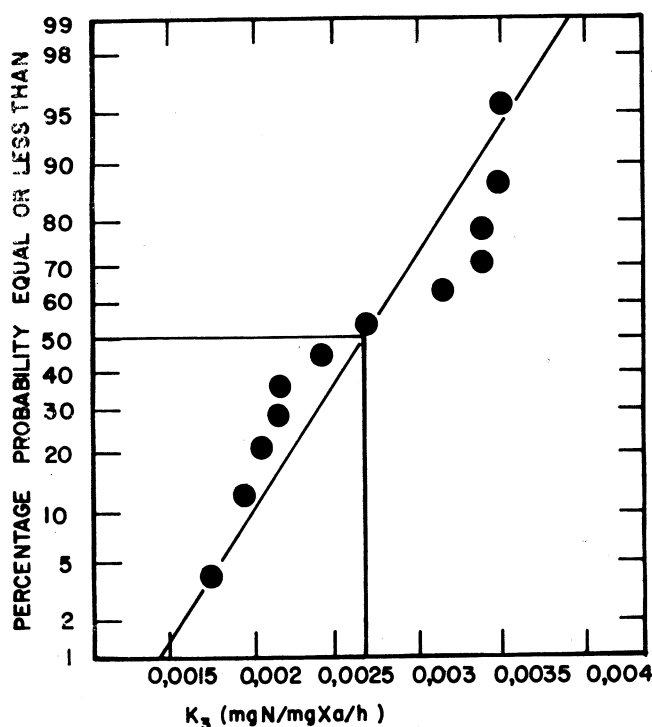


FIGURE 6.8: Statistical plot of the observed denitrification rate constants in a secondary anoxic reactor (K_3).

● Primary anoxic reactor:

Referring to the primary anoxic reactor profiles, two linear reduction phases were observed. The first phase persisted only for a limited period, the second phase persisted for the remaining retention time in the plug flow reactor, see Figure 6.7. To determine the denitrification rate constants of the two phases, it was presumed that the second phase was also operative during the first phase i.e. the slope of the nitrate-time profile of the first phase is the sum of the slopes of the two phases; the slope of the first phase is governed by $(K_1 + K_2)$. Later it will be shown that there is good reason to believe that this is indeed so – see Section 4.1.2 below. The separation of the phases and their associated K values are best appreciated by graphically setting out the nitrate-time profile as shown in Figure 6.9. Hence the system reduction in the pre-denitrification reactor can be written:

$$\begin{aligned} \Delta N_{nps} &= \Delta N_{n1s} + \Delta N_{n2s} \\ &= K_1 X_a t_1 (a + 1) + K_2 X_a R_{ap} (a + 1) \\ &= K_1 X_a t_1 (a + 1) + K_2 X_a R_{np} \end{aligned} \quad (6.11)$$

where

t_1 = duration of the first denitrification phase; subscripts 1 and 2 refer to the first and second phases of denitrification; subscript p refers to the primary anoxic or pre-denitrification reactor; subscripts a and n on the R parameter refer to actual and nominal retention times respectively.

From the many experimental profiles the values of K_1 and K_2 were calculated using the construction set out in Figure 6.9, and Eq. 6.11. The statistical plots of the K_1 and K_2 values at 14°C and 20°C are shown in Figure 6.10. From the mean values at these two temperatures, the following temperature dependency equations were obtained,

$$K_{1T} = 0,720 (1,20)^{(T-20)} \text{ (mgNO}_3\text{-N/mgVASS/d)} \quad (6.12)$$

$$K_{2T} = 0,1008 (1,08)^{(T-20)} \text{ (mgNO}_3\text{-N/mgVASS/d)} \quad (6.13)$$

for $T \geq 13^\circ\text{C}$

$$K_{2T} = K_{3T} \text{ for } 13^\circ\text{C}.$$

The temperature sensitivity of the denitrification rate constants given above and in Eq. 6.10 can be considered reliable only in the range 12°C to 24°C.

From the experimental data as analysed above, Marais and co-workers came to the conclusion that the system reduction of nitrate in the first phase of denitrification, ΔN_{n1s} , was approximately proportional to the influent biodegradable COD concentration, S_{bi} (see Eq. 2.7) i.e.

$$\Delta N_{n1s} = \alpha S_{bi} \quad (6.14)$$

Experimentally, for raw or settled wastewater, the mean value of α was found to be 0,028 mgN/mg biodegradable COD, and this value of α appeared to be independent of temperature. Taking account

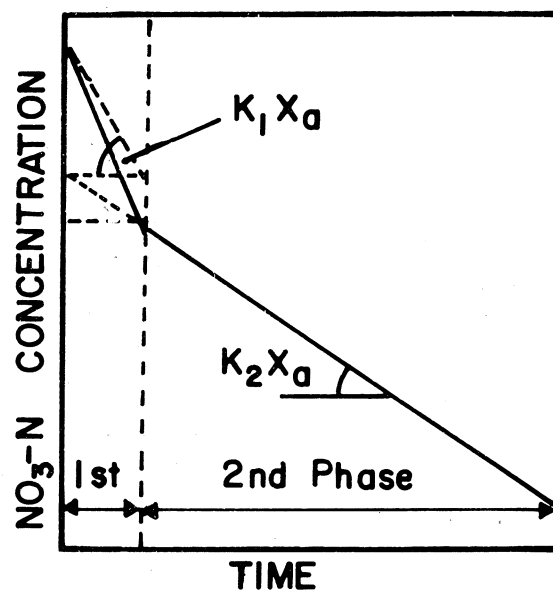


FIGURE 6.9: Denitrification rate constants K_1 and K_2 in a primary anoxic reactor.

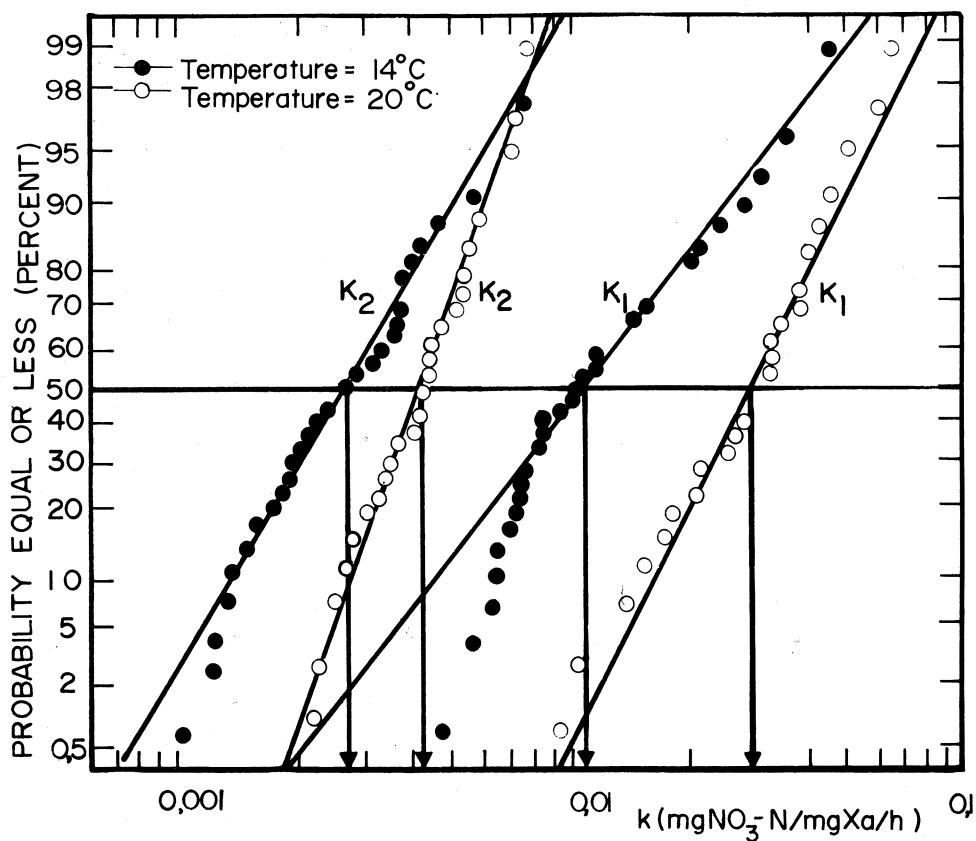


FIGURE 6.10: Statistical plots of denitrification rate constants K_1 and K_2 (pre-denitrification) at 14°C and 20°C .

of these characteristics, the system nitrate reduction in the primary anoxic reactor can be written as

$$\Delta N_{nps} = \alpha S_{bi} + K_2 X_a R_{np} \quad (6.15)$$

and the duration of the first phase of denitrification can be found from Eq. 6.11 as

$$t_1 = \alpha S_{bi} / \{K_1 X_a (1 + a)\} \quad (6.16)$$

From Eq. 6.16, the duration of the first phase of denitrification in the primary anoxic reactor, t_1 , depends on the recycle ratio and the active mass concentration. Usually t_1 lies in the range 2 to 10 minutes.

- (iii) *Recycle ratios:* The total system nitrate reduction of the primary and secondary anoxic reactors is the sum of Eqs. 6.9 and 6.15 i.e.

$$\Delta N_{nts} = \alpha S_{bi} + K_2 X_a R_{np} + K_3 X_a R_{ns} \quad (6.17)$$

where

ΔN_{nts} = total system nitrate removal.

An important point to note is that Eq. 6.17 is valid only if the nitrate is not reduced to zero in any one of the anoxic reactors. Where this condition is satisfied, then from Eq. 6.17 the magnitude of the recycle ratio does not affect the system reduction. However, there must be a minimum recycle; this is determined by the recycle value that *just* causes the nitrate to be zero at the end of a plug flow reactor. If the recycle is less, the nitrate will become zero before the end of the reactor is reached – the recycle can be increased (with concomitant increase in the nitrate removal) to the point where nitrate *just* starts to appear in the effluent. Once nitrate appears in the effluent from all the anoxic reactors, a further increase in the recycle will have no effect on the system nitrate reduction.

- (iv) *Sludge age:* Marais and co-workers found experimentally that at a specific temperature the value of the respective denitrification constants appeared to remain unchanged irrespective of the sludge age for sludge ages between 10 and 25 days.

4.1.2 Nitrate removal and substrate utilization

It has been established in the biological growth kinetic theory that for the utilization of 1 mgCOD under aerobic conditions, an amount of $(1 - f_{cv} Y_h)$ mg oxygen is required. Numerically with $f_{cv} = 1.48$ mgCOD/mgVSS and $Y_h = 0.45$ mgVSS/mgCOD, the oxygen consumption per substrate utilization is 0.33 mg/mgCOD (Chapter 1, Section 4). In an anoxic reactor with substrate utilisation, nitrate is reduced instead of oxygen. As the oxygen equivalent of nitrate is 2.86 mgO/mgN (from Eq. 6.1), the nitrate consumption per mgCOD utilised is $(1 - f_{cv} Y_h) / 2.86 = 0.116$ mgN/mgCOD or conversely 8.6 mgCOD are required to reduce 1 mgN nitrate. From Eq. 6.14 the nitrate concentration removed during the first phase of denitrification is αS_{bi} where $\alpha = 0.028$ and S_{bi} = biodegradable influent COD concentration. Thus the COD reduction during the first phase of denitrifica-

tion is $0.028 \cdot 8.6 \cdot S_{bi} = 0.24 S_{bi}$. From measurements of the readily biodegradable COD in the influent feed, it was found from Eq. 2.8 that the readily biodegradable COD fraction f_{bs} of raw wastewater was 0.24. It was therefore hypothesized that the rapid first phase of denitrification is associated with the utilization of the readily biodegradable COD of the influent. It was also hypothesized that the second slower denitrification rate in the primary anoxic reactor and the rate in the secondary anoxic reactor are associated with the utilization of the slowly biodegradable particulate COD. From a simulation study of the denitrification behaviour in primary and secondary plug flow anoxic reactors, it was concluded that both these hypotheses appear to be valid. Details of this work (see van Haandel and Marais, 1981) are not important here, however, the following conclusions are: (i) the utilization of readily biodegradable COD and slowly biodegradable COD take place simultaneously (see Section 4.1.1 and Figure 6.9) and (ii) the second rate in the primary anoxic reactor (K_2) and the rate in the secondary anoxic reactor (K_3) both arise from the utilization of adsorbed slowly biodegradable COD but the K_2 is more rapid than K_3 because the concentration of adsorbed slowly biodegradable COD in the primary anoxic reactor is higher than that in the secondary anoxic reactor – in the primary anoxic reactor the adsorbed slowly biodegradable COD arises from that in the influent and organism death whereas that in the secondary anoxic reactor arises from organism death only.

4.1.3 Completely mixed anoxic reactors

Up to this stage, only plug flow reactors have been considered. Although this flow regime is important from a research point of view, it has little practical application in full scale nitrification-denitrification plants. Because the denitrification reaction in an anoxic reactor is approximately zero order with respect to nitrate, the reduction achieved in a reactor is virtually independent of the flow regime. Consequently, the equation developed for the nitrate reduction, Eq. 6.17 should be valid also for completely mixed reactors. Where the plug flow reactors were replaced by completely mixed reactors of the same volume, experimental data indicate that the reductions are substantially the same.

Generally, completely mixed reactors are not useful to obtain basic time behaviour characteristics as it is possible only to measure influent and effluent concentrations. However, if the reduction is a single zero order reaction (as in the case of the secondary anoxic reactor), the correct K value can be calculated from the influent and effluent nitrate concentrations (provided the latter is not zero) but if the reduction is due to two zero order reactions (as in the case of the primary anoxic reactor), the correct K values cannot be separated out; the combined effect of the two K values is to give rise to an apparent K value that varies with the retention time. This point is mentioned because in many instances the literature gives primary anoxic denitrification rate constants calculated from influent and effluent nitrate concentrations from completely mixed primary anoxic reactors. A further cause of error is that the denitrification rate constants in the literature often are

given in terms of mg nitrate removed per mg *volatile suspended solids* per unit time. Constants reported in terms of total VSS should be used with caution because the denitrification rates are first order with respect to the *active* volatile suspended solids (X_a) (see Eq. 6.3), not first order with respect to the total volatile suspended solids (X_v) – the inert fraction of the VSS varies significantly with sludge age and wastewater type (see Figure 4.2), so that the denitrification rate constants in terms of total VSS cannot be applied to wastewaters and sludge ages different to those on which the rates were measured. However, if the wastewater characteristics, sludge age and temperature are known then, via the activated sludge theory given in Chapter 4, the denitrification rate constant, if given in terms of VSS, may be corrected to be in terms of the active volatile solids concentration.

4.2 Cyclic flow and load conditions

When denitrification experiments were run under cyclic loading conditions, the approach outlined above for calculating the extent of denitrification no longer appeared to be applicable – the total nitrate concentration removed was less than observed under steady state loading conditions, the removal of nitrate decreased as the severity of the cyclic load variation increased, and the TKN in the effluent showed a marked cyclic response, being high during the high loading phase and low during the low loading phase, whereas the nitrate

behaved in exactly the opposite fashion. It was evident that to establish a general denitrification model, an alternative approach to that established for analysing the steady state behaviour was needed. This was sought in the general model for the activated sludge process developed by Ekama and Marais (1978). The details of this research work are not essential for using this monograph, but it can be mentioned that a large measure of success was achieved in predicting the behaviour of denitrification under dynamic loading conditions once a number of modifications to the model of Ekama and Marais (1978) were accepted. Interested readers are referred to the following publications: Dold, Ekama and Marais (1980); Ekama, van Haandel and Marais (1979); Siebritz, Ekama and Marais (1980); van Haandel and Marais (1981); van Haandel, Ekama and Marais (1981) and van Haandel, Dold and Marais (1982).

The outcome of the research work described in the references above, was a general dynamic model of the activated sludge process including nitrification and denitrification. The theoretical predictions of the general model correlated well with the experimental data observed under both constant and cyclic loading conditions at laboratory and pilot scale.

The general dynamic model, which describes the activated sludge process in basic biological mechanisms, was utilised to study the validity of the empirical denitrification constants K . A comparison of the theoretically predicted and experimentally observed K values is given in Figures 6.11 and 6.12. These figures

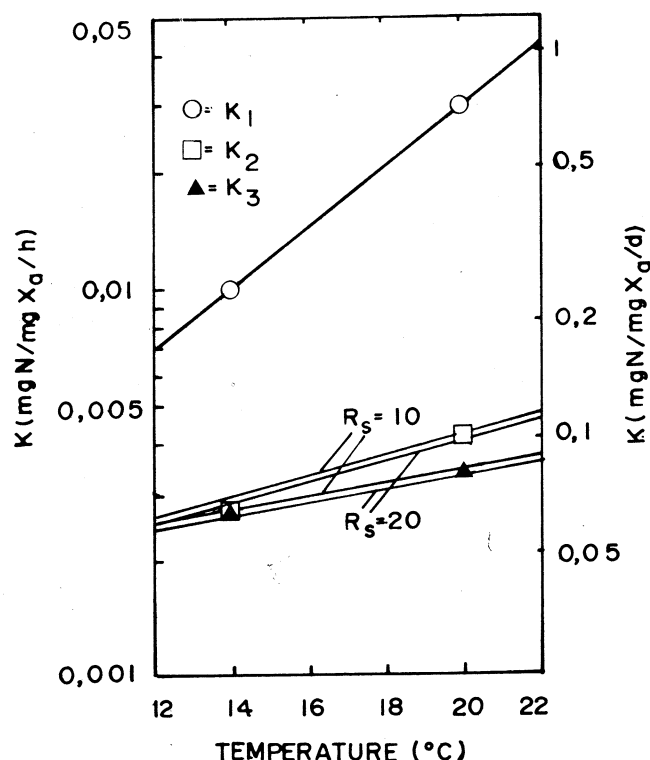


FIGURE 6.11: Simulated and experimental denitrification rate constants K_1 , K_2 and K_3 versus temperature for 10 and 20 days sludge age.

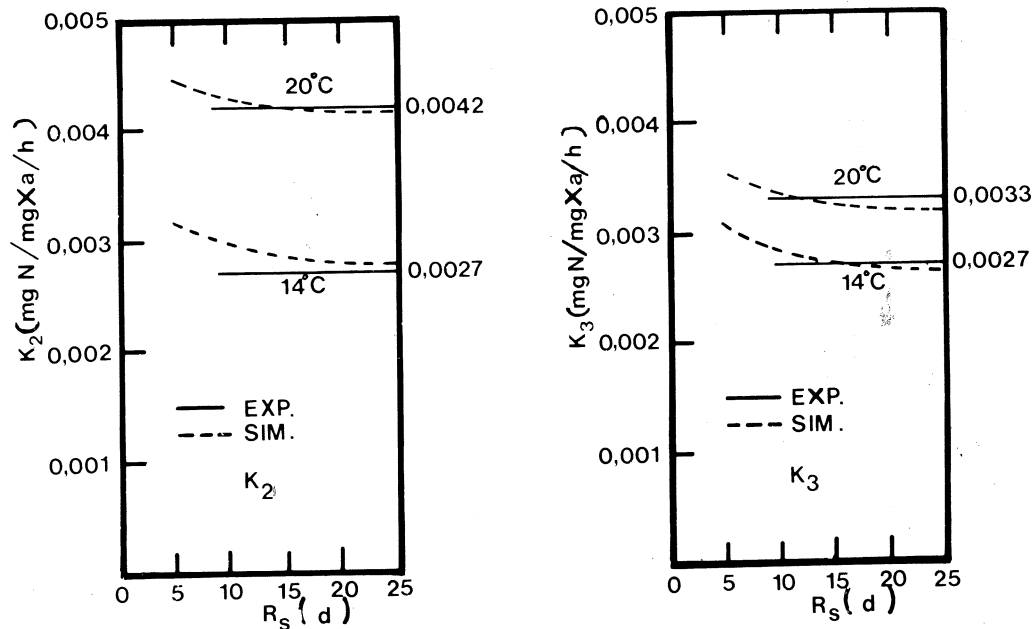


FIGURE 6.12: Simulated and experimental denitrification rate constants K_2 and K_3 versus sludge age at 14°C and 20°C.

show that for sludge ages between 10 and 30 days and temperatures between 12 and 22°C, good correlation between simulated and experimental K values is obtained. The good correlation indicates that although the K values are empirical and have no fundamental biological kinetic basis (of the kind on which the general dynamic model is based), they nevertheless appear to have such consistency that they can be applied to the design of anoxic reactors for constant loading conditions.

5. DENITRIFICATION POTENTIAL

From the empirical denitrification equation of the primary and secondary anoxic reactors, Eqs. 6.15 and 6.9, it is clear that there is a maximum concentration of nitrate that can be removed by the anoxic reactors. In terms of these equations, this maximum for the primary anoxic reactor depends on the influent biodegradable COD, the active mass concentration and the retention time, and for the secondary anoxic reactor only on the active mass concentration and the retention time. The maximum concentration of nitrate that can be removed by an anoxic reactor is known as the *denitrification potential* of the reactor.

5.1 Denitrification potential of the primary anoxic reactor

From Eq. 6.15, the denitrification potential of the primary anoxic reactor, D_{p1} is given by

$$D_{p1} = \alpha S_{bi} + K_2 X_a R_{np} \quad (6.18)$$

Now $X_a R_{np} = X_a V_a / Q$ (where V_a = volume of the primary anoxic reactor and Q = daily average influent flow). The term $X_a V_a$ represents the mass of sludge in the primary anoxic reactor, which can be expressed as a fraction of

the total mass of sludge in the process. Let f_{x1} be the mass of sludge in the primary anoxic reactor as a fraction of the total mass of sludge in the process i.e. the primary anoxic sludge mass fraction, then from Eq. 4.10

$$\begin{aligned} X_a V_a &= f_{x1} M(X_a) \\ &= f_{x1} Y_h R_s Q S_{bi} / (1 + b_h R_s) \end{aligned}$$

Dividing through by Q

$$X_a V_a / Q = f_{x1} Y_h R_s S_{bi} / (1 + b_h R_s) \quad (6.19)$$

Substituting the RHS of Eq. 6.19 for $X_a R_{np}$ into Eq. 6.18 yields

$$D_{p1} = S_{bi} \{ \alpha + K_2 f_{x1} Y_h R_s / (1 + b_h R_s) \} \quad (6.20)$$

where

- D_{p1} = denitrification potential of the primary anoxic reactor (mgN/l influent)
- S_{bi} = biodegradable COD concentration of the influent
- α = fraction of nitrate removed by the initial rapid phase of denitrification
- $\alpha = f_{bs} (1 - f_{cv} Y_h) / 2.86$ (see Section 4.1.2 above)

where

- f_{bs} = readily biodegradable fraction of the influent biodegradable COD
- f_{x1} = primary anoxic sludge mass fraction.

In Eq. 6.20, it is assumed that the initial rapid rate of denitrification is always complete, i.e. the actual retention time in the primary anoxic reactor is always longer than the time required to utilize all the readily biodegradable COD in the influent, given by t_1 in Eq. 6.16. If t_1 is taken to be the minimum actual hydraulic retention time, it can be shown by substituting Eq. 4.10

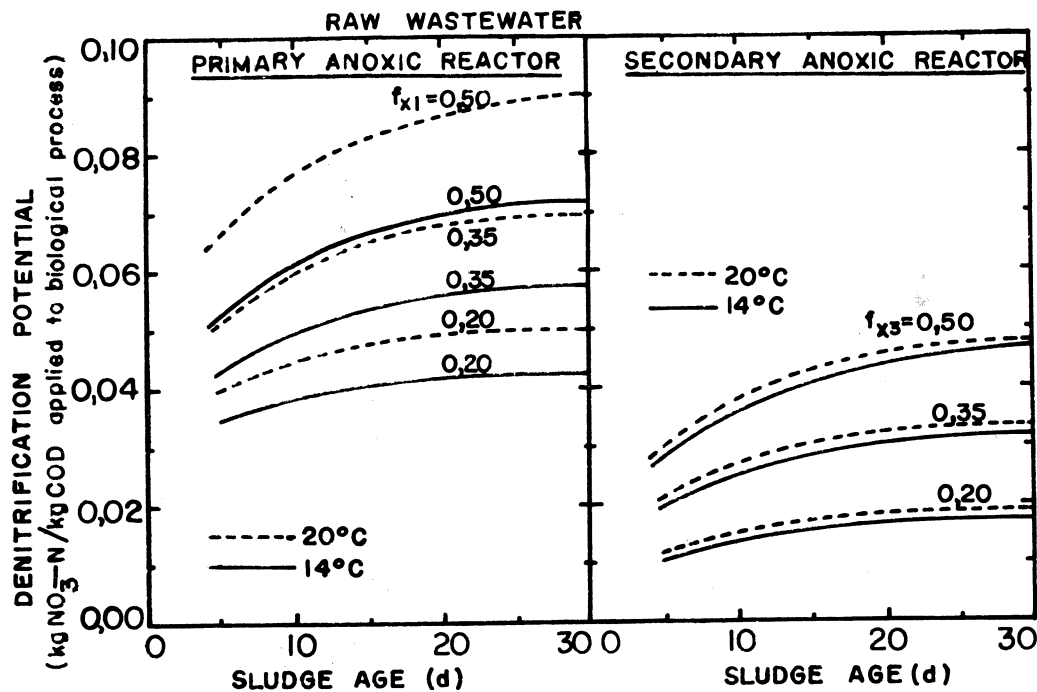


FIGURE 6.13: Denitrification potential per kgCOD applied to biological process versus sludge age for primary (left) and secondary (right) anoxic reactors of different unaerated sludge mass fractions (f_{x1} and f_{x3}) for raw wastewater at 14°C and 20°C.

for X_a and Eq. 6.16 for t_1 , that the minimum primary anoxic sludge mass fraction f_{x1min} to deplete the readily biodegradable COD is

$$f_{x1min} = \frac{f_{bs}(1 - f_{cv}Y_h)(1 + b_hR_s)}{2,86 K_1 Y_h R_s} \quad (6.21)$$

Substituting the values of the kinetic constants into Eq. 6.21, $f_{x1min} < 0,08$ for $R_s > 15$ days at 14°C. This value is much lower than most practical primary anoxic reactors so that Eq. 6.20 will be valid in most cases.

A diagrammatic representation of Eq. 6.20 is given in Figure 6.13 for different primary anoxic sludge mass fractions (f_{x1}) and temperatures. The diagram shows that the denitrification potential of a primary anoxic reactor increases as sludge age, anoxic sludge mass fraction and temperature increase, but that the denitrification potential is most sensitive to the anoxic sludge mass fraction (f_{x1}).

5.2 Denitrification potential of the secondary anoxic reactor

The denitrification potential of the secondary anoxic reactor is found in a similar fashion as that for the primary anoxic reactor and is found to be given by

$$D_{p3} = S_{bi} f_{x3} K_3 Y_h R_s / (1 + b_{hT} R_s) \quad (6.22)$$

where

D_{p3} = denitrification potential of the secondary anoxic reactor (mgN/l influent)

f_{x3} = anoxic sludge mass fraction of the secondary anoxic reactor.

A diagrammatic representation of Eq. 6.22 is given in Figure 6.13 for different anoxic sludge mass fractions and temperatures. The denitrification potential increases as the sludge ages, temperature and anoxic sludge mass fraction (f_{x3}) increase, but is most sensitive to the anoxic sludge mass fraction (f_{x3}).

6. DESIGN PROCEDURE FOR N REMOVAL PROCESSES

In the preceding sections, four important parameters were defined and discussed. These are (1) maximum unaerated sludge mass fraction (f_{xm} , Chapter 5, Section 4.4), (2) nitrification capacity (N_c , Chapter 5, Section 5.3), (3) denitrification potential (Section 5 above) and the TKN/COD ratio (Chapter 3, Section 3.3). These parameters greatly facilitate the design of a nitrogen removal process. The salient aspects of these four parameters are briefly set out below.

(i) Maximum unaerated sludge mass fraction

As nitrification is a prerequisite for denitrification the fraction of the sludge mass under aerobic conditions must be sufficiently large to allow complete nitrification. At a particular sludge age, this sets a limit on the fraction of sludge that is not aerated i.e. sets a maximum unaerated sludge mass fraction (f_{xm}). This maximum unaerated sludge mass fraction is dependent principally on the maximum specific growth rate of the nitrifiers at 20°C (μ_{nm20})

attainable with the wastewater, the minimum average temperature of the wastewater, ($T_{\min}$) and the sludge age of the process (R_s). In order to ensure nitrification, it is recommended that f_{xm} is limited at 0,50 to 0,60 and that the minimum aerobic sludge mass fraction is increased by a factor of safety of 1,25 to 1,35 (see Chapter 5, Section 4.4).

(ii) Nitrification capacity of process

The nitrification capacity is the concentration of nitrate per unit of influent flow that the process produces by nitrification of the influent TKN concentration. If the unaerated sludge mass fraction satisfies the conditions set out above, complete nitrification is likely to take place and the effluent TKN concentration will be between 2 and 3 mgN/l. Under these conditions the nitrification capacity is independent of the process configuration and is the difference between the influent TKN concentration and the sum of the nitrogen required for sludge production and effluent TKN concentration (Chapter 5, Section 5.3).

(iii) Denitrification potential of process

The denitrification potential of a process is the maximum mass of nitrate per unit influent flow that the process as designed can denitrify. It is directly proportional to the influent biodegradable COD concentration and the position and sizes of the anoxic reactors because denitrification in the primary anoxic reactor is faster than in the secondary anoxic reactor.

(iv) The influent TKN/COD ratio

Because the nitrification capacity and denitrification potential are approximately proportional to the influent TKN and COD concentrations respectively, the influent TKN/COD ratio is a relative measure of the mass of nitrate generated in the process by nitrification and the mass of nitrate that can be removed in the process by denitrification.

6.1 Principles of the design procedure

From the wastewater characteristics i.e. influent TKN and COD concentrations (N_{ti} and S_{ti}), maximum specific growth rate of the nitrifiers at 20°C (μ_{nm20}), the readily biodegradable COD fraction (f_{bs}) and the minimum average wastewater temperature ($T_{\min}$), the maximum unaerated sludge mass fraction (f_{xm}) and the nitrification capacity (N_c) can be calculated for a selected sludge age (R_s). The f_{xm} then can be subdivided into primary and secondary anoxic sludge mass fractions (f_{x1} and f_{x3}) and this division fixes the denitrification potential of these two reactors and hence also of the process. When considering the subdivision of f_{xm} , it should be remembered that the denitrification potential of the primary anoxic reactor (D_{p1}) is greater than that

of the secondary anoxic reactor (D_{p3}) for equal anoxic sludge mass fractions, but that sufficient nitrate must be recycled to the primary reactor to fully utilize its denitrification potential.

The relative sizes of the primary and secondary anoxic sludge mass fractions (f_{x1} and f_{x3}), as well as whether or not a secondary anoxic reactor is to be included in the process configuration, depend on the influent TKN/COD ratio. If the objective is to remove as much nitrogen as possible, the denitrification potential of the process (which may or may not include a secondary anoxic reactor) must be as close as possible to the nitrification capacity. Complete denitrification is possible only when the process configuration incorporates a secondary anoxic reactor and if the sum of the denitrification potentials of the primary and secondary anoxic reactors is greater than the nitrification capacity, i.e. $D_{p1} + D_{p3} > N_c$ (provided D_{p1} is fully utilized by having a sufficiently high mixed liquor a-recycle).

If complete denitrification is not possible due to a high TKN/COD ratio, nitrate will be present in the effluent. If the effluent nitrate concentration is greater than about 5 to 6 mgN/l, the incorporation of a secondary anoxic reactor becomes an inefficient utilization of the unaerated sludge mass fraction; an improved nitrate removal will be obtained if the secondary anoxic reactor (f_{x3}) is incorporated into the primary anoxic reactor (f_{x1}). Removal of the secondary anoxic reactor from the Bardenpho process (Figures 6.5) reduces the configuration to a Modified Ludzack-Ettinger (MLE) process (Figure 6.4).

The above discussion indicates that in order to select a nitrogen removal process configuration, it is necessary to check whether or not complete denitrification can be achieved.

6.2 Can complete denitrification be achieved?

In the nitrogen removal processes, the maximum anoxic sludge mass fraction available for denitrification, $f_{x_{dm}}$, is equal to the maximum unaerated sludge mass fraction f_{xm} , i.e.

$$f_{x_{dm}} = f_{xm} \quad (6.23)$$

where f_{xm} is given by Eq. 5.24 for selected R_s , μ_{nmT} and $T_{\min}$.

To check whether or not complete denitrification can be obtained with the maximum anoxic sludge mass fraction, the following reasoning is applied: In the Bardenpho process (which has the potential for complete denitrification), for a fixed underflow s-recycle ratio,* the mixed liquor a-recycle ratio governs the distribution of the nitrate from the aerobic reactor between the primary and secondary anoxic reactors – the higher the a-recycle ratio with respect to the s-recycle ratio, the greater the proportion of nitrate recycled to the primary anoxic reactor. For the selected a- and s-recycle ratio, the best denitrification performance is obtained when the primary anoxic reactor is just loaded to its denitrification potential. If the primary anoxic reactor is loaded to its denitrification potential, the nitrate concentration in the outflow of the reactor is zero and

*Recycle ratios specified with respect to the daily average influent flow rate.

the nitrate concentration in the aerobic reactor is given by $N_c/(a+s+1)$ i.e. the nitrification capacity diluted into the total flow entering the aerobic reactor. Knowing the nitrate concentration in the aerobic reactor and including the effect of the dissolved oxygen concentration (DO) in the aerobic reactor and underflow s-recycle, the equivalent nitrate load imposed on the primary and secondary anoxic reactors by the selected a- and s-recycles can be calculated. Now a useful analytical device is to increase the equivalent nitrate load on the secondary anoxic reactor by a factor K_{2T}/K_{3T} so that the reactor acts like a primary anoxic reactor except that it receives no readily biodegradable COD. Now adding the equivalent nitrate load on the primary anoxic reactor to the adjusted equivalent nitrate load on the secondary anoxic reactor, a total equivalent nitrate load is obtained. Because the secondary anoxic reactor has been transformed to act like a primary anoxic reactor, their respective contributions to total denitrification potential of the process need not be known, and the maximum anoxic sludge mass fraction (f_{xdm}) can be assumed to be in the form of a primary anoxic reactor. Consequently the process denitrification potential D_{pp} can be calculated by means of Eq. 6.20 with $f_{x1} = f_{xdm}$. Now if the process denitrification potential D_{pp} of the maximum anoxic sludge mass fraction f_{xdm} is greater or equal to the total equivalent nitrate load, complete denitrification is possible for the selected a- and s-recycle ratios. In contrast, if the D_{pp} of f_{xdm} is less than the total equivalent nitrate load, complete denitrification is not possible for the selected a- and s-recycle ratios, and nitrate will be present in the effluent.

Following the above reasoning and assuming the DO concentrations in the a- and s-recycles are O_a and O_s mgO/l respectively, the following equation for the effluent nitrate concentration N_{ne} for the Bardenpho process can be developed:

$$N_{ne} = \frac{\left\{ \frac{N_c}{a+s+1} + \frac{O_a}{2.86} \right\} \left\{ a + \frac{K_{2T}}{K_{3T}} (s+1) \right\} + \frac{s \cdot O_s}{2.86} - D_{pp}}{\left\{ \frac{K_{2T}}{K_{3T}} + s \left(\frac{K_{2T}}{K_{3T}} - 1 \right) \right\}} \quad (6.24)$$

where

- N_{ne} = effluent nitrate concentration (mgN/l)
- N_c = nitrification capacity (mgN/l)
- a, s = mixed liquor and sludge underflow recycle ratios respectively
- K_{2T}, K_{3T} = second denitrification rate in primary anoxic reactor and denitrification rate in secondary anoxic reactor respectively (mgN/mgVASS/d)
- O_a, O_s = dissolved oxygen concentration in mixed liquor a and underflow s recycle respectively (mgO/l)
- D_{pp} = denitrification potential at temperature $T^\circ C$ on the assumption that the maximum anoxic sludge mass fraction f_{xdm} found from Eq. 6.23 is all in the form of a primary anoxic reactor i.e. D_{pp} is given by Eq. 6.20 with $f_{x1} = f_{xdm}$ and K_2 adjusted to $T^\circ C$ with Eq. 6.13. (6.25)

Also, the optimum primary and secondary anoxic sludge mass fractions f_{x1} and f_{x3} , and the total f_{xdt} are given by

$$f_{x1} = \frac{\left\{ \frac{N_c}{a+s+1} + \frac{O_a}{2.86} \right\} a + (N_{ne} + \frac{O_s}{2.86}) s - \frac{f_{bs}(1-f_{cv}Y_h) S_{bi}}{2.86}}{\left\{ S_{bi} \frac{Y_h R_s}{(1+b_{HT}R_s)} K_{2T} \right\}} \quad (6.26)$$

$$f_{x3} = \frac{\left\{ (s+1) \left[\frac{N_c}{a+s+1} + \frac{O_a}{2.86} - N_{ne} \right] \frac{K_{2T}}{K_{3T}} \right\}}{\left\{ S_{bi} \frac{Y_h R_s}{(1+b_{HT}R_s)} K_{2T} \right\}} \quad (6.27)$$

$$f_{xdt} = f_{x1} + f_{x3} \quad (6.28)$$

where

f_{x1}, f_{x3}, f_{xdt} = primary, secondary and total anoxic sludge mass fractions respectively.

For a selected sludge age (R_s) and wastewater characteristics ($S_{ti}, N_{ti}, f_{bs}, \mu_{nm20}, T_{min}, f_{up}, f_{us}$) – from which f_{xm} (Eq. 5.24), f_{xdm} (Eq. 6.23), K_{2T}/K_{3T} (Eqs. 6.13 and 6.10), N_c (Eq. 5.29) and D_{pp} (Eq. 6.25) can be calculated – and selected DO concentrations in the a- and s-recycle ratios (O_a and O_s), the only unknowns in Eqs. 6.24 to 6.28 are the a- and s-recycle ratios. The s-recycle ratio usually is specified (at say 1) to obtain satisfactory settling tank performance. Hence N_{ne}, f_{x1} and f_{x3} all depend on the a-recycle ratio only, and hence these three parameters can be calculated for various choices of the a-recycle ratio. If $N_{ne} \leq 0,0$ mgN/l for some specified s- and selected a-recycle ratios, then complete denitrification is possible at these recycle ratios.

In the solution procedure, if Eq. 6.24 yields $N_{ne} < 0$, N_{ne} must be set equal to zero before substitution into Eqs. 6.26 and 6.27. When N_{ne} has to be set equal to zero, it will be noticed that f_{xdt} obtained from Eq. 6.28 is less than the f_{xdm} assumed to calculate D_{pp} . The reason for this is that when $N_{ne} < 0$, $D_{pp} > N_c$. By inserting $N_{ne} = 0$ in Eqs. 6.26 and 6.27, these equations give f_{x1} and f_{x3} that will just produce complete denitrification for the selected a- and specified s-recycle ratios, so that f_{xdt} turns out to be less than f_{xdm} .

For the specified s-recycle ratio, Eqs. 6.24 to 6.28 are valid only for a-recycle ratios falling between a lower and an upper limit:

- (i) **Lower limit:** the a-recycle must not be lower than that which gives f_{x1} as calculated from Eq. 6.26 less than f_{x1min} as calculated from Eq. 6.21 – if $f_{x1} < f_{x1min}$, the readily biodegradable COD is not completely utilized in the primary anoxic reactor leading to inefficient denitrification and also making Eq. 6.20 for D_{p1} invalid.
- (ii) **Upper limit:** the a-recycle ratio must not be increased above that which gives f_{x3} from Eq. 6.27 a smaller value than that required to remove only the DO discharged to the secondary anoxic reactor, i.e. $D_{p3} \geq (1+s) O_a/2.86$ where D_{p3} is given by Eq. 6.22. Hence the minimum secondary anoxic sludge mass fraction is given by

$$f_{x3min} = \frac{(1+s) O_a (1+b_{HT} R_s)}{2.86 S_{bi} Y_h R_s K_{3T}} \quad (6.29)$$

Only the a-recycle ratios falling between the lower and upper limits specified above are valid a-recycle ratios.

By analysing the process with a specified s-recycle ratio for increasing values of valid a-recycle ratios, then, if $N_{ne} > 0$, N_{ne} and the optimal subdivision of f_{xdm} into f_{x1} and f_{x3} , or, if $N_{ne} \leq 0$, the total anoxic sludge mass fraction to just obtain $N_{ne} = 0$, f_{xdt} , and its optimal subdivision into f_{x1} and f_{x3} can be plotted versus the a-recycle ratio.

For design purposes where the process is to operate between temperatures T_{min} and T_{max} , the following sequence of calculations needs to be made to check whether complete denitrification can be achieved and if so, to select the optimal process configuration:

- Step 1: Select wastewater characteristics; S_{ti} , N_{ti} , f_{bs} , f_{up} , f_{us} , μ_{nm20} , T_{max} , T_{min} .
- Step 2: Select R_s and S_f .
- Step 3: Calculate f_{xm} for T_{min} from Eq. 5.24.
- Step 4: With f_{xm} and R_s calculate S_f for T_{max} from Eq. 5.24.
- Step 5: Estimate N_{te} for T_{max} and T_{min} .
- Step 6: With f_{up} and f_{us} , calculate N_s for T_{max} and T_{min} for selected R_s from Eq. 4.23.
- Step 7: Calculate N_c from Eq. 5.29.
- Step 8: Calculate D_{pp} from Eq. 6.25 for T_{max} and for T_{min} with f_{xdm} given by Eq. 6.23.
- Step 9: Select s , O_a and O_s .
- Step 10: Calculate f_{x1min} from Eq. 6.21 for T_{max} and for T_{min} .
- Step 11: Calculate f_{x3min} from Eq. 6.29 for T_{max} and for T_{min} .
- Step 12: Select the a-recycle ratio and calculate N_{ne} for T_{min} from Eq. 6.24.
- Step 13: If $N_{ne} < 0$, set $N_{ne} = 0$.
- Step 14: With N_{ne} calculate f_{x1} , f_{x3} and f_{xdt} from Eqs. 6.26 to 6.28.
- Step 15: Check that $f_{x1} \geq f_{x1min}$ and $f_{x3} \geq f_{x3min}$. If not, discard selected a-recycle ratio as invalid.
- Step 16: Repeat steps 12 to 15 for different a-recycle ratios.
- Step 17: Repeat steps 12 to 16 for T_{max} .

The analysis as to whether or not complete denitrification can be achieved must be undertaken at the *lowest expected temperature* (T_{min}), because the denitrification potential D_{pp} decreases with decrease in temperature. If complete denitrification can be obtained at T_{min} (i.e. $N_{ne} < 0$ from Eq. 6.24 for some valid a-recycle ratio at the specified s-recycle ratio), then f_{x1} is fixed at the *highest expected temperature* (T_{max}). In repeating the calculation at T_{max} it will be found that f_{xdt} at T_{max} is less than f_{xdt} at T_{min} , and for a certain a-recycle ratio, there will be a minimum f_{xdt}^* . The f_{x1} value corresponding to the minimum f_{xdt} at T_{max} fixes the primary anoxic sludge mass fraction for both T_{max} and T_{min} . Now f_{x3} is fixed at the *minimum expected temperature* (T_{min}) at the f_{x3} value corresponding to the fixed f_{x1} value and the total anoxic sludge mass fraction f_{xdt} for both T_{max} and T_{min} is given by $f_{x1} + f_{x3}$. The f_{x3} is fixed at T_{min} because if it were fixed at T_{max} , then f_{xdt} will be insufficient to achieve complete denitrification at T_{min} because for fixed f_{x1} , f_{xdt} at T_{max} is less than f_{xdt} at T_{min} . With f_{x1} and f_{x3} fixed for both T_{max} and T_{min} , the optimum performance of the process is achieved when it is operated at the a-recycle ratio corresponding to the fixed f_{x1} value, this a-recycle ratio being the optimum a_o . It should be noted that with f_{x1} and f_{x3} fixed,

- (i) a_o decreases with decrease in temperature because D_{p1} decreases with decrease in temperature, and
- (ii) the use of any a-recycle ratio other than a_o results in a poorer denitrification performance; if $a < a_o$, D_{p1} is not fully utilized and if $a > a_o$, unnecessarily large quantities of DO are discharged to the primary anoxic reactor.

In the design of a Bardenpho process for a selected sludge age, if it is found that the total anoxic sludge mass fraction for complete denitrification f_{xdt} is less than the maximum allowed f_{xdm} , the following options are open:

- (1) f_{x1} and f_{x3} can be increased to give an f_{xdt} equal to f_{xdm} . This introduces a factor of safety in denitrification by making an allowance for a reduction in the denitrification rates K at 20°C, a reduction in the influent readily biodegradable COD fraction f_{bs} or minimum temperature (T_{min}) or an increase in influent TKN/COD ratio.
- (2) The sludge age (R_s) can be reduced so that f_{xdm} becomes equal to f_{xdt} . The lower R_s will allow a saving in process volume (see Figure 4.3). The lower R_s can be estimated as follows: Set f_{xdm} equal to the f_{xdt} required for complete denitrification at T_{min} ; determine f_{xm} from Eq. 6.23; with f_{xm} calculate R_s from Eq. 5.24; to check, repeat step by step procedure with the new R_s .

Option (2) is not as good an alternative as option (1) because it excludes a factor of safety on the nitrogen

*If T_{max} is less than about 18°C, a clear minimum cannot be discerned (see Figure 6.15 for 14°C). In this event any reasonable f_{x1} can be selected with the proviso that complete utilization of the readily biodegradable COD can be *assured* even with a significant decrease in K_{120} (see Eqs. 6.12 and 6.21). The existence of a minimum, and the temperatures above which it becomes discernible, depends much on the choice of the DO concentration in the recycles O_a and O_s .

removal. However, exclusion of a factor of safety on denitrification is not critical for N removal in the Bardenpho process because it achieves efficient N removal with respect to the other N removal process (i.e. the MLE process) up to an effluent nitrate concentration of about 5-6 mgN/l. However, the exclusion of a factor of safety on denitrification for the Phoredox process may have serious consequences because the discharge of any nitrate to the anaerobic reactor severely reduces the P removal obtainable in the process. The effect of denitrification efficiency on excess biological P removal is discussed in detail in Chapter 7.

6.3 When complete denitrification cannot be achieved

In the Bardenpho process, if it is found that complete denitrification cannot be obtained, (i.e. $N_{ne} > 0$ from Eq. 6.24 for all valid a-recycle ratios at the specified s-recycle ratios) nitrate will be present in the effluent and f_{xdt} will be equal to f_{xdm} . Generally, if the objective is N removal only, the Bardenpho process will be appropriate only if N_{ne} is less than 5 to 7 mgN/l; when $N_{ne} > 5$ to 7 mgN/l, — which will usually be the case if the TKN/COD ratio $> 0,10$ mgN/mgCOD — the MLE process (Figure 6.4) will yield a better nitrogen removal efficiency for the particular wastewater.

For the selected sludge age (R_s), factor of safety on nitrification (S_f) and wastewater characteristics (S_{ti} , N_{ti} , f_{bs} , T_{min} , T_{max} , f_{up} , f_{us}), the f_{xm} (Eq. 5.24), N_c (Eq. 5.29), f_{xdm} (Eq. 6.23) and D_{pp} (Eq. 6.25 with $f_{x1} = f_{xdm}$) for the MLE process are equal to those of the Bardenpho process and are found by the identical procedure (i.e. steps 1 to 8 given above). For the MLE process, the best denitrification performance is obtained when the anoxic reactor is just loaded with nitrate to its denitrification potential. For a specified s-recycle ratio, the mixed liquor a-recycle ratio which loads the anoxic reactor to its denitrification potential is the optimum a-recycle ratio, a_o , and will yield the minimum effluent nitrate concentration. When the nitrate load imposed on the anoxic reactor by the a- and s-recycles (including DO concentrations in the a- and s-recycles of O_a and O_s mgO/l respectively), is less than or equal to the denitrification potential (D_{pp}), the nitrate concentration in the outflow of the anoxic reactor is zero and that in the aerobic reactor and effluent is $N_c/(a+s+1)$ i.e. the nitrification capacity "diluted" into the total flow entering the aerobic reactor. Equating the nitrate load on the anoxic reactor imposed by the a- and s-recycles (including the DO in the recycles) to the denitrification potential D_{pp} and solving for the a-recycle ratio yields a_o :

$$a_o = \{-B + \sqrt{(B^2 + 4AC)}\} / (2A) \quad (6.30)$$

where

$$A = O_a/2,86$$

$$B = N_c - D_{pp} + \{(s+1)O_a + s.O_s\}/2,86$$

$$C = (s+1)(D_{pp} - sO_s/2,86) - sN_c$$

The effluent nitrate concentration N_{ne} for any

a-recycle ratio less than or equal to a_o is given by Eq. 6.31 but will be a minimum when $a = a_o$, i.e.

$$N_{ne} = N_c/(a+s+1) \text{ mgN/l} \quad (6.31)$$

In the design of an MLE process, it will be found the lower N_c is with respect to D_{pp} (i.e. the lower the influent TKN/COD ratio) the lower the N_{ne} and the higher the a_o . In practice, to operate an MLE process with an a_o -recycle ratio greater than 6:1 will be uneconomical: the increase in nitrate removal by increasing a_o from 6 to 10 is only 5% and, therefore, will not be cost-effective with respect to the increased pumping costs. Hence, the N removal of the MLE process is restricted by a maximum a-recycle ratio of say 6, which results in a minimum N_{ne} of 5 to 7 mgN/l. This restriction on the MLE process will not be a problem in practice because high values for a_o are only obtained for wastewaters with low TKN/COD ratios ($< 0,10$ mgN/mgCOD) and these wastewaters are more effectively treated in a Bardenpho process (see section above). Alternatively, if near-complete or complete N removal is not essential, the sludge age can be reduced; for a fixed TKN/COD ratio this reduces D_{pp} (because f_{xm} is reduced) with respect to N_c and hence reduces a_o . The optimum R_s for the fixed TKN/COD ratio can be found using the procedure set out above with different choices of R_s or, alternatively, the method of van Haandel, Dold and Marais (1982) can be used.

The design procedure for N removal processes set out above will be demonstrated in the next section with the aid of worked examples.

7. DESIGN EXAMPLES

The design of biological N removal plant by means of the procedure set out above is demonstrated by continuing the numerical example given in Chapter 4, Section 12 (in which the design procedure for organic material removal was demonstrated) and Chapter 5, Section 6 (in which the behaviour of nitrification was demonstrated).

7.1 Variability of wastewater characteristics

7.1.1 Variability of nitrification rate

The nitrification rate, as expressed by the maximum specific growth rate of the nitrifiers at 20°C (μ_{nm20}) is strongly dependent on the nature and source of the wastewater, so much so that it is considered a wastewater characteristic. The value can range from 0,65/d in purely domestic wastewaters down to 0,20/d for wastewaters with a large industrial* contribution. Owing to uncertainty in the μ_{nm20} value for a particular wastewater, in the absence of any data on the μ_{nm20} value, a low value, say between 0,3 to 0,4/d, needs to be selected to provide reasonable assurance that a safe design with respect to nitrification will be obtained (see Chapter 5, Section 4.1). In this design example, a value of 0,36/d is selected to demonstrate the situation in a

*The type of industry referred to here is that which produces effluents which inhibit biological activity, e.g. metal finishing industry, plating works, paint production plants etc.

design with a relatively low μ_{nm20} value (see Tables 6.1 and 6.2).

7.1.2 Variability in denitrification rates

The denitrification rates K at 20°C are relatively insensitive to the nature and source of the wastewater. However, different values have been observed with different wastewaters – in instances where the wastewater contains an appreciable industrial* contribution, reduced denitrification rates may be encountered. The denitrification rates used in this design example are the mean values found in the experimental work at the University of Cape Town, i.e. $K_{120} = 0,720$, $K_{220} = 0,101$ and $K_{320} = 0,072$ mgNO₃-N/(mgVASS.d) (see Table 6.2). These values were found on Strandfontein wastewater, which is an approximately average wastewater with a relative small industrial contribution. For most normal municipal wastewaters encountered in South Africa the denitrification rates given above are likely to result in a sufficiently accurate estimate of the denitrification potential of the process. Where there is doubt regarding the applicability of these rates, it is recom-

mended that they are reduced by a factor of safety on denitrification, i.e. by about 1,1 to 1,2 times. The magnitude of the factor of safety should reflect the uncertainty in the applicability of the average denitrification rates given in this monograph.

7.1.3 Variability in readily biodegradable COD fraction

The readily biodegradable COD fraction of a wastewater can vary considerably depending on the composition and origin of the various contributions that make up the wastewater. It has been found that in approximately normal principally domestic raw wastewater, approximately 20% of the total (or 25% of the biodegradable) COD is readily biodegradable. However, where industrial wastewaters containing a high biodegradable organic material content are discharged to the municipal sewerage system, significantly higher readily biodegradable COD fractions can be encountered. For example, yeast factories, breweries, malting processes and fruit and vegetable juicing and canning industries produce wastewaters with high readily biodegradable COD fractions. Municipal wastewaters containing con-

TABLE 6.1 RAW AND SETTLED WASTEWATER CHARACTERISTICS IMPORTANT FOR DENITRIFICATION (FOR OTHER CHARACTERISTICS SEE TABLES 4.3 AND 5.2)

Parameter	Symbol	Value		Units
		Raw	Settled	
Maximum specific growth rate of nitrifiers at 20°C	μ_{nm20}	0,36	0,36	/d
Denitrification rates at 20°C	K_1	0,720	0,720	} mgNO ₃ -N/mgVASS/d.
	K_2	0,101	0,101	
	K_3	0,072	0,072	
Readily biodegradable COD fraction	f_{bs}	0,24	0,33*	

*Note increase on f_{bs} fraction in settled wastewater (see Section 7.1.3) – based on a 40% total COD decrease and a 10% readily biodegradable COD decrease in the primary settling tanks. (See Chapter 4, Section 12.1 and Table 4.3).

TABLE 6.2 TEMPERATURE ADJUSTMENT OF THE DENITRIFICATION RATES K_1 , K_2 and K_3

Constant	Symbol	20°C	θ	14°C	22°C	Eq. No.
1st Denit. rate in Prim. Anoxic	K_{120}^*	0,720	1,200	0,241	1,036	6.12
2nd Denit. rate in Prim. Anoxic	K_{220}^*	0,101	1,080	0,0636	0,1178	6.13
Denit. rate in Second. Anoxic	K_{320}^*	0,072	1,029	0,0607	0,0762	6.10
Max. Spec. growth rate of Nitrifiers	μ_{nm20}^{**}	0,36	1,123	0,18	0,45	5.15
Endog. Resp. rate for Nitrifiers	b_{n20}^{**}	0,04	1,029	0,034	0,042	5.16

*Units: mgNO₃-N/(mgVASS.d)

**Units: per day

*Industries producing biodegradable wastes e.g. malting factories, breweries, fruit and vegetable processing factories, etc. do not fall in this group as these may promote biological N and P removal. (see Section 7.1.3).

tributions from these sources can have relatively high readily biodegradable COD fractions depending on the magnitude of the industrial discharges.

Many factors can reduce the readily biodegradable COD fraction of a wastewater. For example the discharge of waste biological sludge to the sewerage system for sludge handling at treatment works lower down the sewer line can deplete a wastewater of all the readily biodegradable COD, particularly if the combined flow is lifted by Archimedian screw pumps. Further examples are cited in Chapter 7, Section 9.

The above examples demonstrate that the readily biodegradable COD fraction can differ significantly between different wastewaters. Consequently, it is recommended that whenever possible, the fraction should be measured. The importance of this fraction in biological nitrogen (and phosphorus) removal is such that it is always worth the effort of measuring it.*

It should be noted that the readily biodegradable COD fraction (as defined by Eqs. 2.8 and 2.9) is affected by primary sedimentation. This is because primary sedimentation removes a large fraction of the particulate COD but very little soluble COD. Consequently, the readily biodegradable COD concentrations (S_{bsi}) are approximately the same before and after primary sedimentation but, because the total or biodegradable COD (S_{ti} , S_{bi}) has been significantly reduced in the primary settlers, the readily biodegradable COD fraction (f_{ts} , f_{bs}) of the settled wastewater is increased (see Chapter 2, Section 2.1).

For the design example it will be assumed that the readily biodegradable COD fraction with respect to the biodegradable COD f_{bs} for the raw wastewater is 0,24, i.e. the value found for approximately normal principally domestic raw wastewaters. The readily biodegradable COD fraction (f_{bs}) of the settled wastewater, assuming that primary sedimentation removes 40% of the total COD but only 10% of the readily biodegradable COD is 0,33. (see Table 6.1, Chapter 4, Section 12.1 and Chapter 2, Section 2).

7.2 Effect of temperature

From the kinetics of denitrification discussed above, it is clear that the denitrification rates K_1 , K_2 and K_3 all decrease with decreasing temperature. This, combined with the fact that the maximum specific growth rate of the nitrifiers at 20°C (μ_{nm20}) also decreases with decreasing temperature has the result that the critical condition for the nitrogen removal process is the average *minimum* wastewater temperature. The design completed for the average minimum wastewater temperature (T_{min}) is checked at the average maximum wastewater temperature (T_{max}) because the oxygen demand per unit COD load increases as the temperature increases.

To facilitate the calculations of the design example at $T_{min} = 14^\circ\text{C}$ and $T_{max} = 22^\circ\text{C}$, the denitrification rates K_1 , K_2 and K_3 are adjusted for temperature in Table 6.2. Temperature adjustment of μ_{nm20} and b_n is given in Table 5.3, but for convenience is repeated in Table 6.2.

7.3 Maximum unaerated sludge mass fraction

This aspect is discussed in detail in Chapter 5, Section 4.4 but it is briefly repeated here because sizing of the maximum unaerated sludge mass fraction is one of the most important decisions in the design of a nutrient removal process.

For nitrogen removal processes, the larger the unaerated sludge mass fraction (f_{xt}) the greater the nitrogen removal and the lower the effluent nitrate concentration. Generally at a minimum temperature of 14°C, to obtain an f_{xt} of 0,50 for favourable μ_{nm20} values (0,40/d), a sludge age of 15 to 20 days is required to ensure nitrification ($S_f = 1,25$) (see Eq. 5.24); for less favourable μ_{nm20} values (between 0,3 and 0,4/d), the required sludge age is 20 to 30 days (see Figure 6.14).

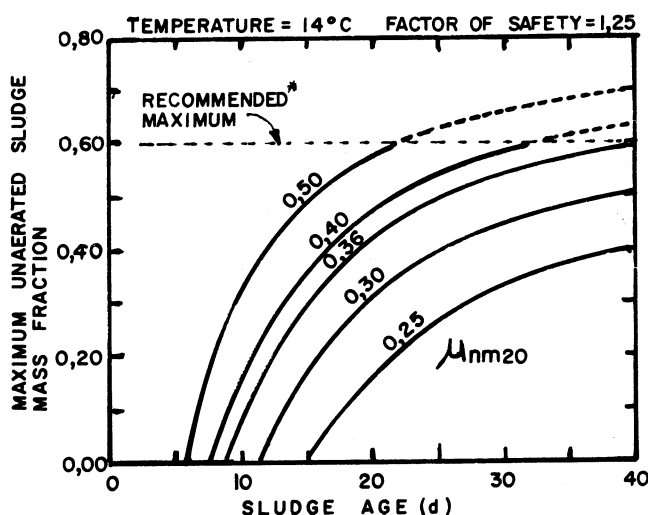


FIGURE 6.14: Maximum unaerated sludge mass fraction versus sludge age for different maximum specific growth rates of the nitrifiers at 20°C (μ_{nm20}) at 14°C and factor of safety (S_f) of 1,25. (From Eq. 5.24).

Now because the volume of the process is directly proportional to the mass of sludge in it (see Chapter 4, Section 6) a 30-day sludge age process will require a process volume about 1/3 larger than a 20 day one. Consequently, for unfavourable μ_{nm20} values it may be preferable to accept a lower f_{xm} and a higher effluent nitrate concentration – a reduction of f_{xm} from 0,50 at $R_s = 30$ d to $f_{xm} = 0,40$ at $R_s = 22$ d results in a 30% saving in process volume and an increase in effluent nitrate concentration of 6 mgN/l (for an influent COD of 600 mg/l and TKN/COD ratio of 0,10). The above argument applies also to favourable μ_{nm20} values because, depending on the effluent quality requirements, it may be acceptable to design a 15-day sludge age process with an effluent nitrate concentration of 5 mgN/l rather than a 20-day process which can achieve complete denitrification (depending on the TKN/COD ratio). Also, for instances where both the

*The readily biodegradable COD fraction f_{ts} (or f_{bs}) or the actual concentration S_{bsi} cannot be measured directly. Its measurement requires the operation and monitoring of a single completely mixed activated sludge process at a very short sludge age under daily cyclic square wave loading conditions (see Appendix 2).

μ_{nm20} and TKN/COD ratio are unfavourable, a 20-day sludge age process with external energy source addition, to augment denitrification, may be a more economical process than a 35-day process. *Clearly, the decision on the unaerated sludge mass fraction and sludge age of the process should not be taken too lightly, because with these two parameters fixed, the N removal of the process with the influent and self-generated energy sources is fixed, so that with fixed influent wastewater characteristics, the effluent quality will be fixed (see Chapter 3).*

For the wastewater characteristics of the design example i.e. $T = 14^\circ\text{C}$ and $\mu_{nm20} = 0,36/\text{d}$ (see Tables 4.3 and 5.2) and accepting a factor of safety on nitrification $S_f = 1,25$, a 25-day sludge age process is required for $f_{xm} = 0,50$ (see Chapter 5, Section 4.4 and Figure 6.14). As this sludge age is not excessively long for $f_{xm} = 0,50$, a 25-day sludge age process will be accepted for a preliminary design analysis.

7.4 Nitrification

Because a factor of safety of 1,25 was imposed in the design of the unaerated sludge mass fraction and sludge age, the minimum sludge age for nitrification is sufficiently below the actual sludge age to assure that complete nitrification will take place i.e. the effluent TKN concentration can be assumed to be 3 mgN/l. (see Chapter 5, Section 4.4). This can be checked by repeating the nitrification calculations given in Chapter 5 for 25 days sludge age.

For the raw wastewater characteristics (i.e. $f_{up} = 0,13$ mgCOD/mgCOD, $f_{us} = 0,05$ mgCOD/mgCOD,

$T_{min} = 14^\circ\text{C}$, $S_{fi} = 600$ mgCOD/l – see Tables 4.3 and 5.2) and 25 days sludge age, and accepting the nitrogen content of the volatile solids f_n to be 0,10 mgN/mgVSS, the nitrogen required for sludge production is given by Eq. 4.23 i.e.

$$N_s = 600 \cdot 0,10 \left[\frac{0,45 (1 - 0,05 - 0,13)}{(1 + 0,20 \cdot 25)} + \frac{0,13}{1,48} \right] \\ = 600 \cdot 0,021 = 12,6 \text{ mgN/l.}$$

The nitrification capacity (N_c) is found from Eq. (5.29) i.e. for the raw wastewater at 14°C

$$N_c = 48 - 12,6 - 3 = 32,4 \text{ mgN/l.}$$

The nitrification oxygen demand is found from Eq. 5.39 i.e.

$$M(O_n) = 4,57 \cdot N_c \cdot Q \\ = 4,57 \cdot 32,4 \cdot 13,33 \cdot 10^6 \text{ mgO/d} \\ = 1973 \text{ kgO/d.}$$

The above calculations for N_s , N_c and $M(O_n)$, including those for the raw wastewater at 22°C and the settled wastewater at 14°C and 22°C , are tabulated in Table 6.3.

In the design, because it is intended to reduce the nitrate concentration as much as possible, the Alkalinity change in the wastewater will be minimized: Assuming that 80% of the nitrate formed is denitrified, the alkalinity change $\Delta\text{Alk} = 7,14 \cdot N_c - 3,57 \cdot (\text{Nitrate denitrified}) = -7,14 \cdot 32,4 + 3,57 \cdot 0,80 \cdot 32,4 = -139 \text{ mg/l as CaCO}_3$.

TABLE 6.3 SUMMARY OF DESIGN CALCULATIONS FOR THE BARDENPHO AND MLE PROCESSES TREATING RAW OR SETTLED WASTEWATER AT 25 DAYS SLUDGE AGE. WASTEWATER CHARACTERISTICS GIVEN IN TABLES 4.3, 5.2 AND 6.1

Parameter	Symbol	Units	Raw Wastewater				Settled Wastewater			
			Bardenpho		MLE		Bardenpho		MLE	
Temperature	T	$^\circ\text{C}$	14	22	14	22	14	22	14	22
Safety factor	S_f		1,25	2,7	1,25	2,7	1,25	2,7	1,25	2,7
Max. unaerated sludge mass	f_{xm}		0,50	0,50	0,50	0,50	0,50	0,50	0,50	0,50
Effluent TKN	N_{te}	mgN/l	3,0	2,0	3,0	2,0	3,0	2,0	3,0	2,0
N for sludge production	N_s	mgN/l	12,6	12,0	12,6	12,0	5,7	5,4	5,7	5,4
Nitrification capacity	N_c	mgN/l	32,4	34,0	32,4	34,0	32,3	33,6	32,3	33,6
Nitrification oxygen demand	$M(O_n)$	kgO/d	1973	2071	1973	2071	1961	2046	1961	2046
Max. anoxic sludge mass	f_{xdm}		0,50	0,50	0,50	0,50	0,50	0,50	0,50	0,50
Max. denitrification potential	D_{pp}	mgN/l	43,1	58,7	43,1	58,7	31,1	41,2	31,1	41,2
DO in a-recycle	O_a	mgO/l	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
DO in s-recycle	O_s	mgN/l	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
Underflow recycle ratio	s		1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
Min. Primary anoxic	f_{x1min}		0,062	0,017	—	—	0,085	0,029	—	—
Min. Secondary anoxic	f_{x3min}		0,024	0,023	—	—	0,039	0,036	—	—
Optimum a-recycle ratio	a_o		3,3	5,0	17,2*	36,0*	2,2	5,0	4,7	13,4*
Effluent nitrate	N_{ne}	mgN/l	0,0	0,0	1,7*	0,9*	4,9	1,7	4,8	2,2*
Primary anoxic	f_{x1}		0,16	0,16	0,50	0,50	0,30	0,30	0,50	0,50
Secondary anoxic	f_{x3}		0,24	0,19	—	—	0,20	0,20	—	—
Total anoxic	f_{xdt}		0,40	0,35	0,50	0,50	0,50	0,50	0,50	0,50

*These are the theoretical results from Eqs. 6.30 and 6.31. Owing to an economical upper limit to the mixed liquor a-recycle ratio, these theoretical a_o values should not be implemented in practice. (see text).

With an influent Alk of 200 mg/l as CaCO₃ (see Table 5.2) the effluent Alk = 200 - 139 = 61 mg/l as CaCO₃, which, from Figure 5.6, will maintain a pH above 7, (see Chapter 5, Section 4.3).

7.5 Denitrification

Accepting a sludge age of 25 days, which allows a maximum unaerated sludge mass fraction f_{xm} of 0,50, the denitrification behaviour of the Bardenpho and MLE processes will be demonstrated for the typical raw and settled wastewaters at 14°C and 22°C.

In the calculations it will be assumed that there is a dissolved oxygen concentration of 2 mgO/l in the interconnecting flows from the aerobic to the anoxic reactors (i.e. $O_a = 2$ mgO/l) and 1 mgO/l in the underflow recycle from the secondary settling tank (i.e. $O_s = 1$ mgO/l). The underflow recycle ratio (s) is assumed to be 1:1.*

For nitrogen removal processes, the maximum anoxic sludge mass fraction ($f_{x_{dm}}$) is equal to the maximum unaerated sludge mass fraction (f_{xm}) i.e. $f_{x_{dm}} = 0,50$ (see Eq. 6.23). The denitrification potential of the primary anoxic reactor at 14°C with $f_{x1} = f_{x_{dm}} = 0,50$ for the raw wastewater is found from Eq. 6.20 (see Eq. 6.25)

$$\begin{aligned} D_{pp} &= (1 - 0,05 - 0,13) 600 \left\{ \frac{0,24 (1 - 0,45 \cdot 1,48)}{2,86} + \right. \\ &\quad \left. 0,0636 \cdot 0,50 \cdot 0,45 \cdot 25 / (1 + 0,20 \cdot 25) \right\} \\ &= 492(0,028 + 0,0596) \\ &= 13,8 + 29,3 \\ &= 43,1 \text{ mgN/l.} \end{aligned}$$

The D_{pp} values for the raw wastewater at 22°C and those for the settled wastewater at 14°C and 22°C are listed in Table 6.3.

The minimum primary and secondary anoxic sludge mass fractions (f_{x1min} and f_{x3min}) are found from Eq. 6.21 and Eq. 6.29 respectively. For example for the raw wastewater at 14°C,

$$\begin{aligned} f_{x1min} &= \frac{0,24(1 - 0,45 \cdot 1,48)(1 + 0,20 \cdot 25)}{2,86 \cdot 0,241 \cdot 0,45 \cdot 25} \\ &= 0,062 \text{ and} \\ f_{x3min} &= \frac{(1 + 1)2,0(1 + 0,20 \cdot 25)}{(1 - 0,05 - 0,13)600 \cdot 0,45 \cdot 25 \cdot 0,0607 \cdot 2,86} \\ &= 0,025 \end{aligned}$$

The values for the raw wastewater at 22°C and those for the settled wastewater at 14°C and 22°C are given in Table 6.3.

At this stage of the design procedure, we have col-

lected all the information necessary to check whether or not complete denitrification can be achieved i.e. steps 1 to 11 of the step by step procedure have been completed.

Continuing the procedure with step 12, a value for the a-recycle ratio is selected and the effluent nitrate concentration is calculated from Eq. 6.24. Selecting $a = 3,3$ then for the raw wastewater at 14°C,

$$\begin{aligned} N_{ne} &= \frac{\left\{ \frac{32,4}{3,3+1+1} + \frac{2}{2,86} \right\} \left\{ 3,3 + \frac{0,0636}{0,0607} (1+1) \right\} + \frac{1 \cdot 1}{2,86} - 43,1}{\left\{ \frac{0,0636}{0,0607} + 1 \left(\frac{0,0636}{0,0607} - 1 \right) \right\}} \\ &= -5,47 \text{ mgN/l} \end{aligned}$$

Because $N_{ne} < 0,0$, N_{ne} must be set equal to zero. Setting $N_{ne} = 0$ and substituting into Eqs. 6.26 to 6.28 yields the optimum primary (f_{x1}), secondary (f_{x3}) and total (f_{xdt}) anoxic sludge mass fractions just to achieve complete denitrification with an a-recycle ratio of 3,3 i.e.

$$\begin{aligned} f_{x1} &= \frac{\left\{ \frac{32,4}{3,3+1+1} + \frac{2}{2,86} \right\} 3,3 + \left(0 + \frac{1}{2,86} \right) 1 - \frac{0,24 \cdot 0,334 \cdot 492}{2,86}}{492 \frac{0,45 \cdot 25}{(1 + 0,20 \cdot 25)} 0,0636} \\ &= 0,16 \end{aligned}$$

$$\begin{aligned} f_{x3} &= \frac{\left\{ (1+1) \left[\frac{32,4}{3,3+1+1} + \frac{2}{2,86} \right] - 0 \right\} \frac{0,0636}{0,0607}}{492 \frac{0,45 \cdot 25}{1 + 0,20 \cdot 25} 0,0636} \\ &= 0,24 \end{aligned}$$

$$f_{xdt} = 0,40$$

The above calculation is repeated for different values of a. The results of N_{ne} , f_{x1} , f_{x3} and f_{xdt} are shown plotted versus a in Figure 6.15a. Repeating the design procedure at 22°C (steps 12 to 15) for the raw wastewater produces the results shown in Figure 6.15b (see also Table 6.3).

Figure 6.15 shows that, for the raw wastewater, complete denitrification can be achieved at 14°C for any a-recycle ratio > 1,8 (Figure 6.15a). At 22°C (Figure 6.15b), a minimum f_{xdt}^{**} for complete denitrification of 0,35 occurs at $a = 5,0$. The f_{x1} corresponding to $a = 5,0$ is 0,16 and this fixes the design value of f_{x1} for both 14°C and 22°C. Now if f_{x3} is fixed at 22°C at the value corresponding to f_{x1} i.e. $f_{x3} = 0,19$ (Figure 6.15b), then f_{xdt} will be insufficient to achieve complete denitrification at 14°C; for a fixed f_{x1} , f_{xdt} at 14°C is greater than f_{xdt} at 22°C (compare Figures 6.15a and b). Hence f_{x3} is fixed at 14°C at the value corresponding to $f_{x1} = 0,16$ i.e. $f_{x3} = 0,24$ (Figure 6.15a). Hence $f_{xdt} = 0,40$ for complete denitrification at 14°C and 22°C.

With f_{x1} and f_{x3} fixed for both 14°C and 22°C, the optimum performance of the process is achieved when it is operated at the a-recycle ratio corresponding to the

*This is usually fixed at a value such that satisfactory settling tank operation is obtained. Secondary settling tank behaviour is described in Chapter 8.

**If T_{max} is less than 18°C, a clear minimum cannot be discerned (see Figure 6.15a at 14°C). In this event, any reasonable f_{x1} can be selected with the proviso that complete utilization of the readily biodegradable COD can be assured even with a significant decrease in K_{120} (see Eqs. 6.12 and 6.21). The existence of a minimum, and the temperature above which it becomes discernible, depends much on the choice of the DO concentrations in the recycles, O_a and O_s .

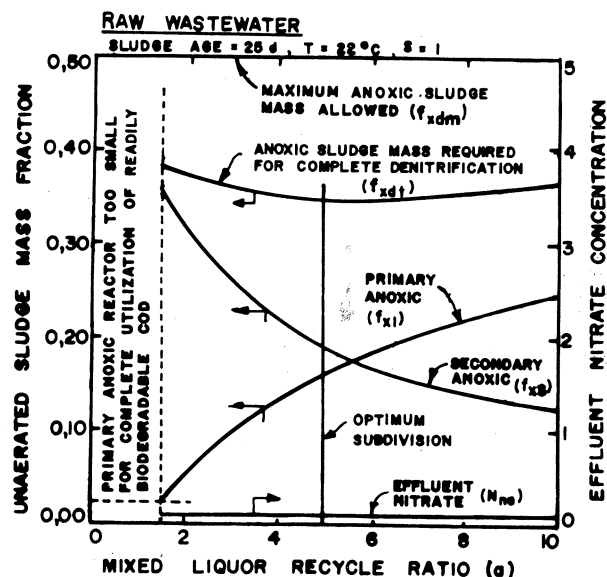
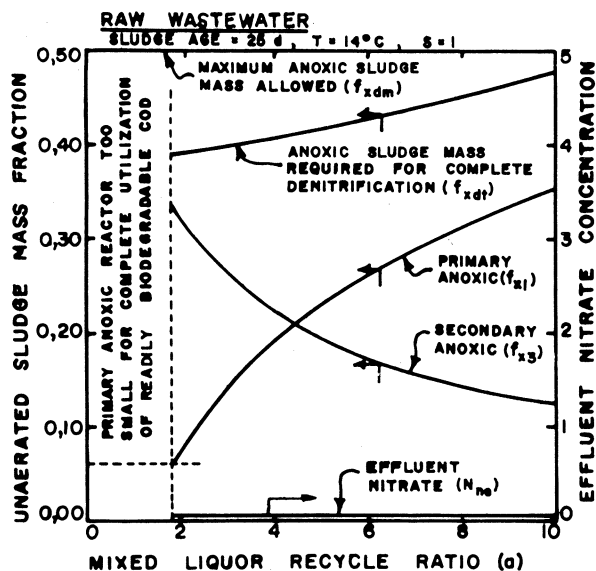


FIGURE 6.15: Primary (f_{x1}), secondary (f_{x3}), total (f_{xdt}) and maximum allowable (f_{xdm}) anoxic sludge mass fractions and effluent nitrate concentrations (N_{ne}) versus mixed liquor a-recycle ratio for the Bardenpho process treating raw wastewater at 25 days sludge age at 14°C (left, Figure 6.15a) and 22°C (right, Figure 6.15b).

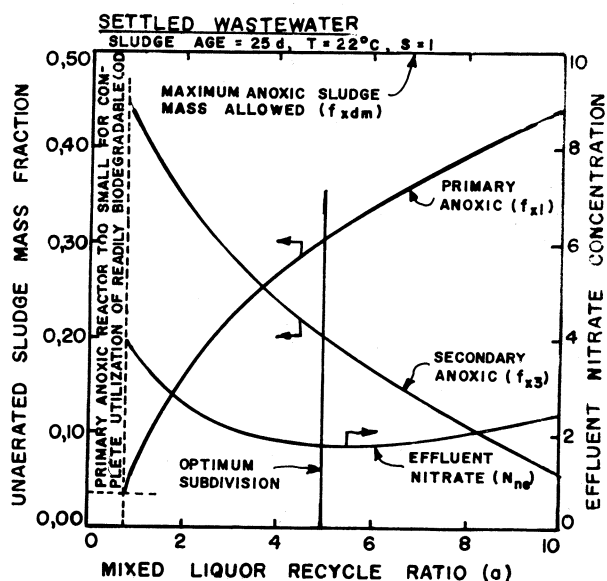
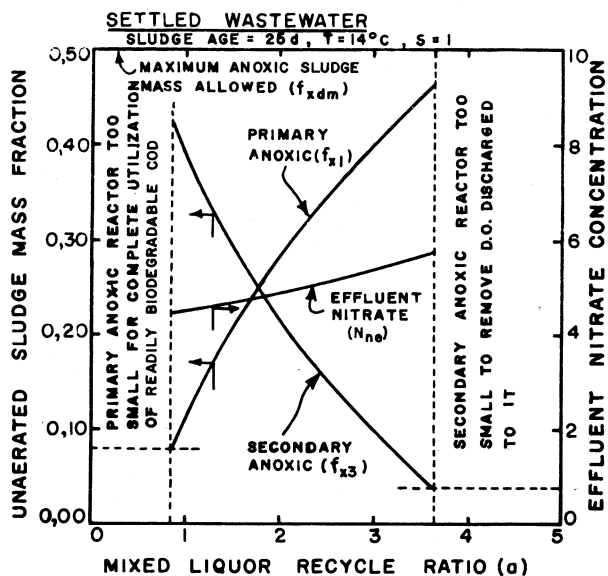


FIGURE 6.16: Primary (f_{x1}), secondary (f_{x3}), total (f_{xdt}) and maximum allowable (f_{xdm}) anoxic sludge mass fractions and effluent nitrate concentration (N_{ne}) for the Bardenpho process treating settled wastewater at 25 days sludge age at 14°C (left, Figure 6.16a) and 22°C (right, Figure 6.16b).

fixed f_{x1} value, this a-recycle ratio being the optimum a_o ; at 14°C, $a_o = 3,3$ (Figure 6.15a) and for 22°C, $a_o = 5,0$ (Figure 6.15b). This demonstrates that as the temperature increases, the a-recycle ratio needs to be increased to maintain optimum performance. This is because with increase in temperature, the denitrification potential of the primary anoxic reactor (D_{p1}) increases and consequently more nitrate must be recycled to it to fully utilize its D_{p1} . It should be noted that with f_{x1} and f_{x3} fixed, the use of any a-recycle ratio other than a_o results in poorer denitrification performance: if $a < a_o$, D_{p1} is not fully utilized and if $a > a_o$, unnecessarily large quantities of DO are discharged to the primary anoxic reactor.

The design parameters calculated above are summarized in Table 6.3.

The design procedure for the settled wastewater at 14°C and 22°C was repeated and the results are shown plotted in Figures 6.16a and b respectively. At 22°C (Figure 6.16b), a minimum effluent nitrate concentration* of 1,7 mgN/l is obtained with $a = 5,0$. The f_{x1} corresponding to $a = 5,0$ is 0,30 and this value fixes f_{x1} . Now because the maximum anoxic sludge mass fraction is fully utilized i.e. $f_{xdt} = f_{xdm}$, f_{x3} is fixed at the difference between f_{xdt} and f_{x1} i.e. $f_{x3} = 0,20$. Hence f_{x1} and f_{x3} are fixed at 0,30 and 0,20 respectively for 14°C and 22°C. The optimum a-recycle ratio is 5,0 at 22°C yielding an effluent nitrate concentration of 1,7 mgN/l and that at 14°C is 2,2 (i.e. the value corresponding to $f_{x1} = 0,30$, Figure 6.16a) yielding $N_{ne} = 4,9$ mgN/l. The design results are listed in Table 6.3.

The design calculations above show that with the raw wastewater, complete denitrification can be obtained and is achieved with a total anoxic sludge mass fraction (f_{xdt}) of 0,40 whereas 0,50 is allowed (f_{xdm}). As N removal only is important in this example, the Bardenpho process is appropriate and (i) if a factor of safety for complete denitrification is required, option 1 (see Section 6.2 above) can be taken, i.e. f_{x1} and f_{x3} are increased such that $f_{x1} + f_{x3} = f_{xdt} = f_{xm} = 0,50$; or (ii) if no factor of safety for complete denitrification is required, option 2 can be taken, i.e. the sludge age can be reduced such that $f_{xm} = f_{xdt} = 0,40$.

For the settled wastewater, complete denitrification cannot be achieved at both 14°C and 22°C. Because the maximum anoxic sludge mass fraction is fully utilized (i.e. $f_{xdt} = f_{xdm} = 0,50$), the denitrification efficiency can be improved only by increasing the sludge age of the process. However, the increase in sludge age is at the expense of an increased process volume. A sludge age of 40 d, which allows $f_{xm} = f_{xdm} = 0,60$,** is required to produce complete denitrification at 14°C (found by trial and error using design procedure). From Eq. 4.21 an increase in sludge age from 25 to 40 days represents an increase in process volume of about 40%.

Such a large increase in volume may not be merited for an improvement in effluent nitrate concentration from 5 to 0 mgN/l. Consequently a sludge age of 25 to 30 d is considered the maximum economical sludge age beyond which there is little cost effective improvement in denitrification efficiency. If complete denitrification must be achieved, it may be preferable to treat raw wastewater at 20-day sludge age which allows an $f_{xdm} = f_{xm} = 0,40$ (see Figure 6.15a). Although this increases the process volume by 80% it removes the problem of primary sludge disposal. Alternatively, when treating settled wastewater, and complete denitrification must be achieved, a 20-day sludge age process with readily biodegradable COD addition to the primary or secondary anoxic reactor may be economically viable – if 60*** mgCOD/l influent readily biodegradable COD is added to the plant, complete denitrification will be obtained with settled wastewater at 14°C and 20-day sludge age. For the design flow of 13,3 Ml/d, the COD addition is $60 \cdot 13,3 = 800$ kg/d i.e. 10% of the raw wastewater COD load.

If complete denitrification is not essential, the Bardenpho process as designed for the settled wastewater i.e. sludge age = 25 d, is appropriate. Alternatively, the feasibility of an MLE process can be checked to see whether or not this process may yield a more economical design. The design procedure for the MLE process is demonstrated below.

The design procedure for the MLE process is identical to that for the Bardenpho process up to step 9 of the step by step procedure. For the same design conditions of the Bardenpho process (i.e. $R_s = 25$ d etc), the results for the MLE process are listed in Table 6.3 and can be seen to be identical to those of the Bardenpho process. Substituting the values for the nitrification capacity N_c and denitrification potential D_{pp} into Eqs. 6.30 and 6.31, the optimum mixed liquor a-recycle ratio a_o and minimum effluent nitrate concentration N_{ne} are obtained; e.g. for the settled wastewater at 14°C –

$$A = 2/2,86 = 0,70$$

$$B = 32,3 - 31,1 + \{(1+1)2 + 1,1\}/2,86 = +2,95$$

$$C = (1+1)(31,1 - 1,1/2,86) - 1,32,3 = +29,2$$

Hence

$$a_o = \{-2,95 + \sqrt{2,95^2 + 4 \cdot 0,70 \cdot 29,2}\} / (2 \cdot 0,70) \\ = 4,7 \text{ and}$$

$$N_{ne} = 32,3 / (4,7 + 1 + 1) = 4,8 \text{ mgN/l.}$$

The above results, as well as those for the settled wastewater at 22°C and those for the raw wastewater at 14°C are listed in Table 6.3.

*If T_{max} is less than 18°C, a clear minimum cannot be discerned (see Figure 6.16b at 14°C). In this event, any reasonable f_{x1} can be selected with the proviso that complete utilization of the readily biodegradable COD can be assured even with a significant decrease in K_{120} (see Eqs. 6.12 and 6.21). The existence of a minimum, and the temperature above which it becomes discernible, depends much on the choice of the DO concentrations in the recycles, O_a and O_s .

**This is the recommended upper limit for the maximum unaerated sludge mass fraction (see Chapter 5, Section 4.4).

***Assuming that $N_{ne} = 7,0$ mgN/l at $R_s = 20$ d (i.e. 2 mgN/l greater than at $R_s = 25$ d), the COD consumption for its removal is $7,0 \cdot 8,6 = 60$ mgCOD/l influent.

The results show that for all cases except one (i.e. for settled wastewater at 14°C), the optimum mixed liquor a-recycle ratio a_o exceeds 5. Although these results include the discharge of dissolved oxygen to the anoxic reactor, recycle ratios above 4 to 5 are not cost effective – the small decrease in effluent nitrate concentration (N_{ne}) which an increase in a-recycle ratio from 5 to 10 produces does not warrant the additional pumping costs (e.g. for $N_c = 33$, $N_{ne} = 4,7$ for $a = 5$ and 2,7 for $a = 10$). Consequently, the a-recycle ratio is limited at 5, which results in a minimum N_{ne} from the MLE process of about 5 to 6 mgN/l for most municipal wastewaters. Hence for N removal processes, if effluent nitrate concentrations lower than 5 to 6 mgN/l are required, the Bardenpho process is the indicated process because this process can achieve low N_{ne} with low a-recycle ratios.

Comparing the Bardenpho and MLE processes, (see Table 6.3) if the effluent quality standards require that complete or near complete denitrification must be produced, then the Bardenpho process is the more superior process for both the raw and settled wastewaters. However, the process needs to be operated at a sludge age of about 20 days for the *raw wastewater* (yielding complete denitrification) and at least 30 days for the *settled wastewater* (yielding an effluent nitrate concentration of about 3 mgN/l) – the settled wastewater process will have a smaller volume than the raw wastewater process but will have the additional problem of primary sludge disposal. Alternatively, if complete denitrification is not necessary and an effluent nitrate concentration of 5 to 6 mgN/l is acceptable, the MLE process will be an acceptable process. For the raw wastewater, a sludge age of only 15 days (yielding $f_{xm} = f_{xdm} = 0,32$) is sufficient to produce an effluent nitrate concentration of about 5 mgN/l with the MLE process and for the settled wastewater, a sludge age of 25 days will yield an effluent nitrate concentration of about 5 mgN/l.

7.6 Process volume and oxygen demand

7.6.1 Process volume

Having determined the subdivision of the sludge mass into anoxic and aerobic fractions to achieve the required nitrogen removal, the actual sludge mass in the process needs to be calculated to determine the volumes of the different reactors. The mass of total (MLSS) or volatile (MLVSS) sludge in the activated sludge process for a selected sludge age and wastewater characteristics is given by Eq. 4.21 (see Chapter 4, Section 6) i.e. for the given raw and settled wastewater characteristics, the MLSS sludge mass in the process at 25-day sludge age and 14°C is 54 200 and 21 850 kgTSS respectively. Selecting an MLSS concentration in the process of 4 000 mg/l* (i.e. 4 kg/m³), the volume of the process treating raw wastewater is 13 550 m³ and that treating settled wastewater is 5 460 m³. Now, because the

sludge mass in the nitrogen removal processes is uniformly distributed in the process, i.e. each reactor has the same MLSS concentration, the volume fractions of the reactors are equal to the sludge mass fractions. Accepting the Bardenpho process design for the raw and settled wastewaters (see Table 6.3) the volume of the reactors are found directly from the sludge mass fraction and total process volumes; reactor retention times, nominal and actual are found from the reactor volumes and the nominal and total flows passing through them – see Table 6.4. Note that the reactor nominal retention time is a consequence of the mass of sludge generated from the influent COD load, the selected MLSS concentration and the sludge mass fraction – the retention time *per se* has no significance in kinetics of nitrification and denitrification (see Chapter 4, Sections 6 and 12).

TABLE 6.4 REACTOR VOLUMES, NOMINAL AND ACTUAL RETENTION TIMES FOR THE BARDENPHO PROCESS TREATING THE TYPICAL RAW AND SETTLED WASTEWATER AT 25-DAY SLUDGE AGE. FOR PROCESS PERFORMANCE SEE TABLE 6.3

Process Parameter	Symbol	Raw Wastewater	Settled Wastewater	Units
Sludge mass	$M X_t$	54 200	21 850	kgTSS
MLSS concentration	X_t	4 000	4 000	mg/l
Total volume	V_p	13 550	5 460	m ³
Total flow	Q	13 330	13 330	m ³ /d
Total nominal retention time	R_{nt}	24,4	9,8	h
Mixed liquor recycle (14°C)	a	3,3	2,2	
Und. recycle	s	1,0	1,0	
Primary anoxic				
Mass fraction	f_{x1}	0,20	0,30	
Volume		2 710	1 640	m ³
Nom. ret. time		4,9	2,9	h
Actual ret. time		1,13	0,92	h
Main aeration				
Mass fraction*		0,45	0,45	
Volume		6 100	2 460	m ³
Nom. ret. time		11,0	4,4	h
Actual ret. time		2,6	1,4	h
Secondary anoxic				
Mass fraction	f_{x3}	0,30	0,20	
Volume		4 060	1 090	m ³
Nom. ret. time		7,3	2,0	h
Actual ret. time		3,7	1,0	h
Reaeration				
Mass fraction*		0,05	0,05	
Volume		680	270	m ³
Nom. ret. time		1,2	0,50	h
Actual ret. time		0,6	0,25	h

*The reaeration reactor is usually taken to have a sludge mass fraction of around 0,05 to 0,07 and this fraction is deducted from the aerobic sludge mass fraction; the remaining aerobic sludge mass fraction constitutes the main aeration sludge mass fraction.

*A method for estimating the reactor MLSS concentration for design conditions is as follows: for a selected X_t calculate the process volume requirements. For accepted settling characteristics of the sludge, calculate the required settler surface area for the selected X_t (see Chapter 8). Estimate the combined cost of the process and settling tanks. It will be found that as X_t increases the cost of the process decreases but the cost of the settling tanks increases. The X_t that gives the minimum combined cost is the indicated MLSS concentration (usually between 3 000 and 6 000 mgTSS/l with settled wastewater being lower than raw wastewater).

7.6.2 Daily average total oxygen demand

The total oxygen demand in a nitrogen removal process is the sum of that required for carbonaceous material (COD) degradation and nitrification, less than recovered by denitrification. The daily average oxygen demand for (i) carbonaceous material removal (MO_c) is given by Eq. 4.15 (see Chapter 4, Section 7), and (ii) nitrification is given by Eq. 5.39 (see Section 7.4 above and Chapter 5, Section 6.5) and these oxygen demands in the Bardenpho process at 25-day sludge age for the typical raw and settled wastewaters at 14°C and 22°C are listed in Table 6.5.

TABLE 6.5 DAILY AVERAGE OXYGEN DEMAND (IN kgO/d) IN THE 25-DAY SLUDGE AGE BARDENPHO PROCESS FOR THE TYPICAL RAW AND SETTLED WASTEWATERS AT 14°C AND 22°C

Oxygen Demand kgO/d	Symbol	Raw		Settled	
		14°C	22°C	14°C	22°C
1. Carbonaceous*	$M(O_c)$	+ 5105	+ 5205	+ 3287	+ 3352
2. Nitrification**	$M(O_n)$	+ 1973	+ 2071	- 1961	+ 2046
3. Denitrification***	$M(O_d)$	- 1235	- 1296	- 1048	- 1216
4. Total	$M(O_t)$	+ 5843	+ 5980	+ 4200	+ 4185

*See Chapter 4, Section 12.3, Eq. 4.27

**See Section 7.4 above

***From Eq. 6.32 and Table 6.3

The oxygen recovered by denitrification (MO_d) is given by 2,86 times the nitrate mass denitrified (see Section 1.3) where nitrate mass denitrified is the product of the daily average influent flow Q and the nitrate concentration denitrified. The nitrate concentration denitrified is given by the difference in the nitrification capacity N_c and the effluent nitrate concentration: Hence

$$M(O_d) = 2,86 \cdot (N_c - N_{ne})Q \quad (6.32)$$

From the denitrification performance data of the Bardenpho process in Table 6.3, the oxygen recovered by denitrification for the typical raw and settled wastewaters are listed in Table 6.5.

For the raw wastewater, Table 6.5 shows that (i) the nitrification oxygen demand (MO_n) is about 40% that required for COD removal (MO_c), (ii) about 60% of MO_n can be recovered by incorporating denitrification, (iii) the additional oxygen demand by incorporating nitrification and denitrification is only 15% of that required for COD removal only and (iv) the effect of temperature on the total oxygen demand is marginal – less than 3% (see also Figure 6.1).

For the settled wastewater, Table 6.5 shows that (i) the nitrification oxygen demand is about 60% of that required for COD removal, (ii) about 50% of the nitrification oxygen demand can be recovered by denitrification, (iii) the additional oxygen demand by incorporating nitrification and denitrification is about 30% of that required for COD removal only and (iv) the effect of

temperature on the total oxygen demand is marginal – less than 3% more at the lower temperature.

Comparing the oxygen demand for the raw and settled wastewaters, it can be seen that the total oxygen demand for the latter is about 30% less than that of the former. This saving is possible because primary sedimentation removes 35 to 45% of the raw wastewater COD (see Chapter 4, Section 12 and Chapter 5, Section 6.5). Furthermore, for the settled wastewater, the nitrification oxygen demand is a greater proportion of the total, and also, less of the nitrification oxygen demand can be recovered by denitrification compared to the raw wastewater. These effects are due to the higher TKN/COD ratio of the settled wastewater.

Generally for design, preliminary estimates of the total daily average oxygen demand can be roughly approximated with the following:

- (i) including complete nitrification;

$$M(O_t) = \{1 + 5 f_{ns}\} M(O_c) \quad (6.33)$$

- (ii) including complete nitrification, and denitrification to the degree possible depending on the TKN/COD ratio (see Figure 6.17)

$$M(O_t) = \{1 + 2 f_{ns}\} M(O_c) \quad (6.34a)$$

for $f_{ns} < 0,09$ (i.e. effluent nitrate
< 2 mgN/l)

$$= \{1 + 2 f_{ns} [1 + 10(f_{ns} - 0,08)]\} M(O_c) \quad (6.34b)$$

for $f_{ns} > 0,09$ (i.e. effluent nitrate
> 2 mgN/l)

where f_{ns} = influent TKN/COD ratio.

These equations, in conjunction with the carbonaceous oxygen demand $M(O_c)$ equation in terms of the influent COD load (MS_{ti}) (see Chapter 4, Section 7) will allow the calculation of preliminary estimates of the daily average total oxygen demand in nitrogen removal processes between sludge ages of 15 to 30 d for raw and settled wastewaters for known influent COD loads and TKN/COD ratios.

Knowing the average daily total oxygen demand, the peak total oxygen demand can be roughly estimated by means of a simple design rule. From a large number of simulations with the general dynamic model, it was found that provided the factor of safety on nitrification (S_f) is greater than 1,25 to 1,35, the relative amplitude {i.e. (Peak-Average)/Average} of the total oxygen demand is a fraction 0,50 of the relative amplitude of the total oxygen demand potential of the influent COD and TKN load {i.e. $Q(S_{ti} + 4,57 \cdot N_{ti})$ }. For example, with the raw wastewater design, if the peak influent total oxygen demand potential is obtained at a time of day when the influent flow rate, COD and TKN concentrations are 23,1 Ml/d, 863 mgCOD/l and 65 mgN/l respectively – i.e. $23,1 (863 + 4,57 \cdot 65) = 26\,800$ kgO/d – and the average total oxygen demand potential is 13,33 ($600 + 4,57 \cdot 48 = 10\,920$ kgO/d (see Table 4.3), the relative amplitude of the total influent oxygen demand potential is $(26\,800 - 10\,920)/10\,920 = 1,45$; hence the relative amplitude of the total oxygen demand is approximately $0,50 \cdot 1,45 = 0,72$; from Table 6.5 the

average daily total oxygen demand is 5 980 kgO/d (at 22°C) and hence the peak oxygen demand is $1,72 \cdot 5\,980 = 10\,300$ kgO/d. As with all simplified design rules, the above rule should be used with discretion and caution, and where possible, the peak total oxygen demand is best estimated by means of the general dynamic activated sludge model.

7.7 Closure

In the above design example, the raw wastewater TKN/COD ratio is 0,08 mgN/mgCOD. If the calculations are repeated for the raw wastewater but assuming different TKN/COD ratios and sludge ages, it will be found that (i) for a fixed sludge age, once the TKN/COD ratio is above that with which complete denitrification can be achieved, the effluent nitrate concentration increases as the TKN/COD ratio increases and (2) at a fixed TKN/COD ratio above that with which complete denitrification can be achieved, as the sludge age increases, the effluent nitrate concentration decreases. This implies that at a fixed sludge age, the N removal potential of the process is fixed so that the effluent quality depends on the magnitude of the influent TKN/COD ratio; however, the N removal potential of the process may be improved by increasing the sludge age.

Based on 14°C and the raw wastewater characteristics of the design example (except the TKN/COD ratio) the behaviour described above is shown graphically in Figure 6.17, in which the influent TKN/COD ratio is plotted versus the effluent nitrate concentration that can be achieved at different sludge ages. The process volume required at the different sludge ages (as a fraction of that required at 25 d) is also given on Figure 6.17. The areas for which the Bardenpho and MLE process achieve the optimum N removal are demarcated and generally where the effluent nitrate concentration is greater than 5 to 6 mgN/l, the MLE process will yield an improved efficient nitrate concentration at an economical mixed liquor a recycle ratio i.e. $< 4:1$. It should be noted that at a fixed TKN/COD ratio, the total average daily oxygen demand does not vary significantly between different sludge ages so that aeration costs do not significantly affect the conclusions that can be drawn from Figure 6.17 (see Figure 6.1).

Figure 6.17 shows that for a TKN/COD ratio of say 0,095 mgN/mgCOD, complete denitrification can be achieved if the sludge age is 30 days in a Bardenpho process. However, if an effluent quality of 7 mgN/l is acceptable, then this can be achieved in an MLE process at a sludge age of 20 days. The process volume saving that can be made by reducing the effluent nitrate standard from 0 to 6 mgN/l is $1 - 0,83/1,16 = 0,28$ i.e.

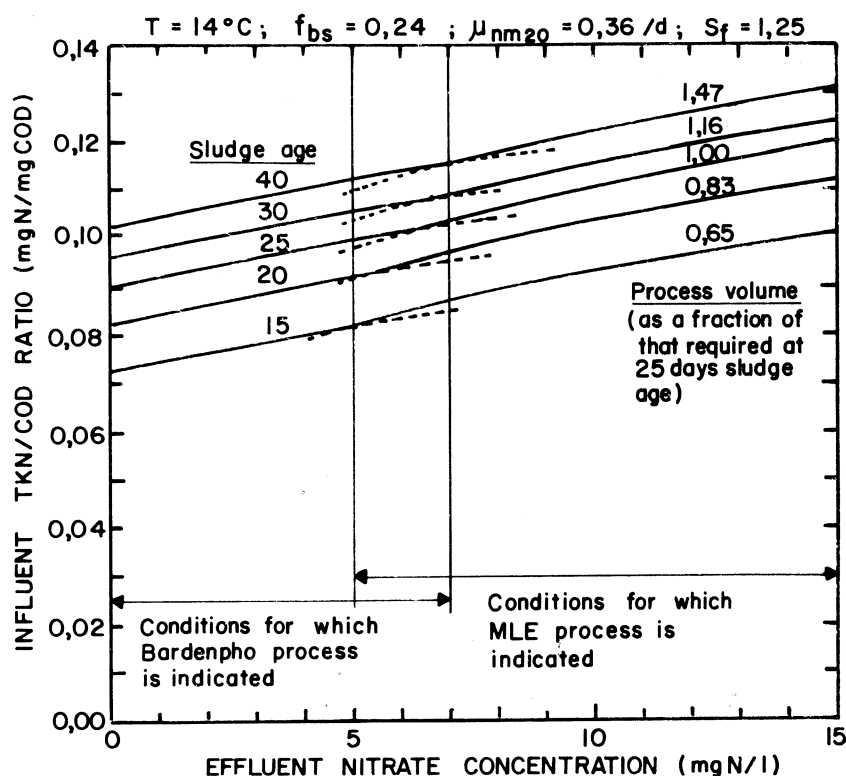


FIGURE 6.17: Effluent nitrate concentration attainable for different influent TKN/COD ratios for N removal processes operating at different sludge ages for $T=14^{\circ}\text{C}$, readily biodegradable COD fraction (f_{bs}) = 0,24 and maximum specific growth rate of the nitrifiers at 20°C (μ_{nm20}) = 0,36/d.

the MLE process at 20-day sludge age with 7 mgN/l is about 30% smaller than the Bardenpho process at 30-day sludge age with complete denitrification. Figure 6.17 shows also that for low μ_{nm20} values, 14°C and 25-day sludge age (to avoid excessively large process volumes), the Bardenpho process can achieve (i) complete denitrification for TKN/COD ratios less than 0,090 mgN/mgCOD; (ii) near complete denitrification (i.e. effluent nitrate concentration between 1 and 5 mgN/l) for TKN/COD ratios between 0,09 and 0,10 mgN/mgCOD and (iii) incomplete denitrification (i.e. effluent nitrate concentration above 5 mgN/l) for TKN/COD ratios above 0,10 mgN/mgCOD; hence the MLE process is the indicated process for TKN/COD ratios above 0,10. The above limits are relatively sensitive to the μ_{nm20} value and readily biodegradable COD fraction; (i) the limits are given for a low μ_{nm20} value (around 0,36/d) because this will be the usual case in design unless sufficient information is available justifying the use of a higher one (see Appendix 3) – if μ_{nm20} is greater than 0,40/d, the results will be similar to those shown in Figure 6.17 except that they can be achieved at shorter sludge ages; (ii) the limits given are for an approximately normal raw wastewater readily biodegradable COD fraction f_{bs} of 0,24 – there are many factors which can affect the f_{bs} fraction of a wastewater (see Section 7.1.3 above) so that at the design stage, the value of f_{bs} also will be uncertain [unless the value has been measured (see Appendix 2)] and it is likely that a value of 0,24 will be used initially for want of a better estimate; if the actual f_{bs} value is greater than 0,24, lower effluent nitrate concentrations will be obtained than those given in Figure 6.17 and if f_{bs} is lower than 0,24, higher effluent nitrate concentrations will be obtained.

Although the limits given in Figure 6.17 should not be taken as general because of their dependence on the wastewater characteristics, in particular μ_{nm20} and f_{bs} , they are useful for setting approximate guidelines for N removal process selection and the sludge age at which it is to be operated to achieve a certain effluent nitrate concentration for a given influent TKN/COD ratio. The guidelines in Figure 6.17 are not intended to obviate doing detailed design calculations but serve the purpose of demonstrating the behaviour of N removal processes. Indeed it is recommended that in process design, the wastewater characteristics are changed to encompass a range of possible values for the particular wastewater and to undertake design calculation for combinations of different wastewater characteristics. This will allow the selection of a process design that can best deal with anticipated changes in wastewater characteristics and will

expose weaknesses in the design so that control measures can be incorporated to deal with fluctuations in wastewater characteristics during the life of the plant.

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